

Miitjee

Subject : Physics,
Chemistry,
Biology

Standard : 11,12

Total Mark : 720

NEET GRAND Test 2

(Answer Key)

Paper Set : 1

Date : 15-06-2026

Time : 0H:0M

Physics - Section A (MCQ)

1 - A	2 - D	3 - B	4 - A	5 - B	6 - B	7 - A	8 - A	9 - B	10 - B
11 - C	12 - C	13 - C	14 - B	15 - A	16 - A	17 - C	18 - D	19 - D	20 - C
21 - A	22 - C	23 - C	24 - C	25 - C	26 - A	27 - D	28 - B	29 - A	30 - C
31 - C	32 - C	33 - C	34 - B	35 - C	36 - B	37 - C	38 - B	39 - D	40 - B
41 - D	42 - B	43 - A	44 - C	45 - A					

Chemistry - Section A (MCQ)

46 - B	47 - C	48 - A	49 - D	50 - B	51 - D	52 - D	53 - A	54 - B	55 - B
56 - C	57 - B	58 - C	59 - A	60 - B	61 - B	62 - B	63 - B	64 - B	65 - D
66 - D	67 - A	68 - B	69 - B	70 - C	71 - B	72 - A	73 - B	74 - C	75 - C
76 - A	77 - B	78 - C	79 - A	80 - C	81 - D	82 - C	83 - A	84 - B	85 - B
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Biology - Section A (MCQ)

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101 - C	102 - A	103 - D	104 - D	105 - D	106 - C	107 - B	108 - A	109 - A	110 - A
111 - D	112 - B	113 - C	114 - C	115 - B	116 - C	117 - B	118 - C	119 - B	120 - B
121 - A	122 - D	123 - C	124 - A	125 - A	126 - A	127 - B	128 - A	129 - D	130 - D
131 - B	132 - A	133 - B	134 - B	135 - B	136 - A	137 - C	138 - A	139 - C	140 - B
141 - C	142 - D	143 - B	144 - C	145 - D	146 - D	147 - A	148 - D	149 - D	150 - A
151 - D	152 - D	153 - A	154 - B	155 - D	156 - A	157 - D	158 - D	159 - A	160 - C
161 - A	162 - C	163 - D	164 - B	165 - C	166 - A	167 - B	168 - A	169 - B	170 - B
171 - C	172 - C	173 - B	174 - A	175 - C	176 - C	177 - B	178 - B	179 - B	180 - C

Subject : Physics,
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(Solutions)

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Physics - Section A (MCQ)

- (1) A disc of mass m and radius r rolls down an inclined plane of height h . When it reaches the bottom of the plane, its rotational kinetic energy is (g = acceleration due to gravity).

- (A) $\frac{mgh}{3}$ (B) $\frac{mgh}{6}$
(C) $\frac{mgh}{2}$ (D) $\frac{mgh}{4}$

Solution:(Correct Answer:A)

(A) For a disc rolling down an inclined plane, the total potential energy at the top is converted into translational and rotational kinetic energy at the bottom.

By the law of conservation of energy: $mgh = K_{trans} + K_{rot}$.

For a disc, the moment of inertia $I = \frac{1}{2}mr^2$.

Since the disc rolls without slipping, $v = r\omega$, so $\omega = \frac{v}{r}$.

The translational kinetic energy is $K_{trans} = \frac{1}{2}mv^2$.

The rotational kinetic energy is

$$K_{rot} = \frac{1}{2}I\omega^2 = \frac{1}{2}\left(\frac{1}{2}mr^2\right)\left(\frac{v}{r}\right)^2 = \frac{1}{4}mv^2.$$

$$\text{Total energy } mgh = \frac{1}{2}mv^2 + \frac{1}{4}mv^2 = \frac{3}{4}mv^2.$$

$$\text{Thus, } \frac{1}{2}mv^2 = \frac{2}{3}mgh.$$

Substituting this into the expression for rotational kinetic energy:

$$K_{rot} = \frac{1}{4}mv^2 = \frac{1}{2}\left(\frac{1}{2}mv^2\right) = \frac{1}{2}\left(\frac{2}{3}mgh\right) = \frac{mgh}{3}.$$

- (2) Out of the following statements, which one is 'NOT' correct in the case of a thermodynamic process?

- (A) In an isothermal process, $\Delta T = 0$
(B) In an isobaric process, $\Delta P = 0$
(C) In an isochoric process, $W = 0$
(D) In an isothermal process, $Q = 0$

Solution:(Correct Answer:D)

(D) In an isothermal process, the temperature remains constant, so $\Delta T = 0$. However, heat exchange Q is not necessarily zero; it is equal to the work done ($Q = W$).

In an adiabatic process, the heat exchange Q is zero.

Therefore, the statement 'In an isothermal process, $Q = 0$ ' is incorrect.

- (3) The work done in splitting a water drop of radius R into 64 droplets is (T = Surface tension of water). (in πTR^2)

- (A) 6 (B) 12
(C) 8 (D) 24

Solution:(Correct Answer:B)

(B) Given the radius of the large drop is R . Let r be the radius of each smaller droplet. Since the total volume remains constant:

$$\frac{4}{3}\pi R^3 = 64 \times \frac{4}{3}\pi r^3$$

$$\therefore r^3 = \frac{R^3}{64} \Rightarrow r = \frac{R}{4}$$

$$\text{Initial surface energy } E_1 = 4\pi R^2 T.$$

$$\text{Final surface energy } E_2 = 64 \times (4\pi r^2 T) =$$

$$64 \times 4\pi \times \left(\frac{R}{4}\right)^2 \times T = 64 \times 4\pi \times \frac{R^2}{16} \times T = 16\pi R^2 T.$$

$$\text{Work done } W = E_2 - E_1 = 16\pi R^2 T - 4\pi R^2 T = 12\pi R^2 T.$$

- (4) **Assertion:** On a rainy day, it is difficult to drive a car or bus at high speed.

Reason: The value of the coefficient of friction is lowered due to the wetting of the surface.

- (A) If both *Assertion* and *Reason* are correct and the *Reason* is a correct explanation of the *Assertion*.
(B) If both *Assertion* and *Reason* are correct but *Reason* is not a correct explanation of the *Assertion*.
(C) If the *Assertion* is correct but *Reason* is incorrect.
(D) If both the *Assertion* and *Reason* are incorrect.

Solution:(Correct Answer:A)

(A) On a rainy day, the roads become wet.

Water acts as a lubricant between the tires and the road surface, which significantly lowers the coefficient of friction (μ).

Since the frictional force $f = \mu N$ (where N is the normal force), a decrease in μ leads to a decrease in the available frictional force.

This reduced friction makes it difficult to maintain control, increases the braking distance, and increases the chances of skidding at high speeds.

Therefore, both the *Assertion* and *Reason* are correct, and the *Reason* is the correct explanation for the *Assertion*.

- (5) For a satellite moving in an orbit around the earth, the ratio of kinetic energy to potential energy is

- (A) 2 (B) $-1/2$
(C) $1/\sqrt{2}$ (D) $\sqrt{2}$

Solution:(Correct Answer:B)

(B) The kinetic energy (K) of a satellite of mass m orbiting at a distance r from the center of the Earth (mass M) is given by $K = \frac{GMm}{2r}$.

The potential energy (U) of the satellite is given by $U = -\frac{GMm}{r}$.

The ratio of kinetic energy to potential energy is

$$\frac{K}{U} = \frac{GMm/2r}{-GMm/r} = -\frac{1}{2}.$$

Note: The magnitude ratio is $1/2$, but considering the sign, the ratio is $-1/2$.

- (6) The length of the simple pendulum is made 3 times the original length. If T is its original time period, then the new time period will be:

- (A) $3T$ (B) $\sqrt{3}T$
(C) $\frac{T}{\sqrt{3}}$ (D) $\frac{T}{3}$

Solution:(Correct Answer:B)

(B) The time period T of a simple pendulum is given by the formula: $T = 2\pi\sqrt{\frac{L}{g}}$, where L is the length of the pendulum and g is the acceleration due to gravity.

From this formula, we can see that $T \propto \sqrt{L}$.

Let the original length be $L_1 = L$ and the original time period be $T_1 = T$.

The new length is $L_2 = 3L$.

Let the new time period be T_2 .

Using the proportionality $T \propto \sqrt{L}$, we have: $\frac{T_2}{T_1} = \sqrt{\frac{L_2}{L_1}}$.

Substituting the values: $\frac{T_2}{T} = \sqrt{\frac{3L}{L}} = \sqrt{3}$.

Therefore, $T_2 = \sqrt{3}T$.

- (7) Energy is not carried by which of the following waves?

- (A) Stationary
(B) Progressive
(C) Transverse
(D) Electromagnetic

Solution:(Correct Answer:A)

(A) stationary wave (also known as a standing wave) is formed by the superposition of two identical waves traveling in opposite directions.

In a stationary wave, the energy is trapped between the nodes and does not propagate through the medium. While the energy oscillates between kinetic and potential forms at different points, there is no net transport of energy across the medium.

In contrast, progressive waves, transverse waves, and electromagnetic waves are all characterized by the transport of energy from one point to another.

Therefore, the correct answer is A.

- (8) "Heat cannot by itself flow from a body at lower temperature to a body at higher temperature" is a statement or consequence of

- (A) Second law of thermodynamics
(B) Conservation of momentum
(C) Conservation of mass
(D) First law of thermodynamics

Solution:(Correct Answer:A)

(A) The statement provided is known as the Clausius statement of the Second Law of Thermodynamics.

It asserts that it is impossible to construct a device that operates in a cycle and produces no effect other than the transfer of heat from a lower-temperature body to a higher-temperature body.

Therefore, the correct option is A.

- (9) Preeti reached the metro station and found that the escalator was not working. She walked up the stationary escalator in time t_1 . On other days, if she remains stationary on the moving escalator, then the escalator takes her up in time t_2 . The time taken by her to walk up on the moving escalator will be

- (A) $\frac{t_1 t_2}{t_2 - t_1}$ (B) $\frac{t_1 t_2}{t_1 + t_2}$
(C) $t_1 - t_2$ (D) $\frac{t_1 + t_2}{2}$

Solution:(Correct Answer:B)

(B) Let d be the distance of the escalator.

Velocity of Preeti on the stationary escalator is $v_1 = \frac{d}{t_1}$.

Velocity of the moving escalator is $v_2 = \frac{d}{t_2}$.

When Preeti walks on the moving escalator, her net velocity with respect to the ground is $v = v_1 + v_2$.

$$v = \frac{d}{t_1} + \frac{d}{t_2} = d \left(\frac{t_1 + t_2}{t_1 t_2} \right).$$

The time t taken to cover the distance d with net velocity v is:

$$t = \frac{d}{v} = \frac{d}{d \left(\frac{t_1 + t_2}{t_1 t_2} \right)} = \frac{t_1 t_2}{t_1 + t_2}.$$

- (10) 300 g of water at 25°C is added to 100 g of ice at 0°C . The final temperature of the mixture is (in $^\circ\text{C}$)

- (A) 12.5 (B) 0
(C) 25 (D) 50

Solution:(Correct Answer:B)

(B) Heat released by 300 g of water when cooling from 25°C to 0°C is given by $Q_{\text{released}} = m \cdot c \cdot \Delta T$.

$$Q_{\text{released}} = 300 \text{ g} \times 1 \text{ cal/g}^\circ\text{C} \times (25^\circ\text{C} - 0^\circ\text{C}) = 7500 \text{ cal}.$$

Heat required to melt 100 g of ice at 0°C into water at 0°C is given by $Q_{\text{required}} = m \cdot L_f$.

$$Q_{\text{required}} = 100 \text{ g} \times 80 \text{ cal/g} = 8000 \text{ cal}.$$

Since $Q_{\text{required}} > Q_{\text{released}}$, the available heat is insufficient to melt all the ice.

Therefore, the mixture will reach a state of thermal equilibrium at 0°C with some ice remaining unmelted.

The final temperature of the mixture is 0°C .

- (11) A beaker full of hot water is kept in a room. If it cools from 80°C to 75°C in t_1 minutes, from 75°C to 70°C in t_2 minutes, and from 70°C to 65°C in t_3 minutes, then:

- (A) $t_1 = t_2 = t_3$ (B) $t_1 < t_2 = t_3$
(C) $t_1 < t_2 < t_3$ (D) $t_1 > t_2 > t_3$

Solution:(Correct Answer:C)

(C) According to Newton's law of cooling, the rate of cooling is directly proportional to the temperature difference between the body and its surroundings.

Mathematically, $\frac{d\theta}{dt} = k \left(\frac{\theta_1 + \theta_2}{2} - \theta_0 \right)$, where θ_0 is the room temperature.

Since the temperature difference decreases as the water cools, the rate of cooling decreases.

Therefore, the time taken to cool by the same amount (5°C) increases as the temperature of the water approaches the room temperature.

Comparing the mean temperatures:

$$\left(\frac{80+75}{2} \right) > \left(\frac{75+70}{2} \right) > \left(\frac{70+65}{2} \right).$$

Thus, the rate of cooling is highest for the first interval and lowest for the third interval.

Consequently, $t_1 < t_2 < t_3$.

- (12) The length of a metal wire is L_1 when the tension is T_1 and L_2 when the tension is T_2 . The unstretched length of the wire is:

- (A) $\frac{L_1 + L_2}{2}$
(B) $\sqrt{L_1 L_2}$
(C) $\frac{T_2 L_1 - T_1 L_2}{T_2 - T_1}$
(D) $\frac{T_2 L_1 + T_1 L_2}{T_2 + T_1}$

Solution:(Correct Answer:C)

(C) Let the unstretched length of the metal wire be L and its cross-sectional area be A .

According to Hooke's Law, the Young's modulus Y is given by $Y = \frac{T/A}{\Delta L/L}$, where ΔL is the change in length.

For tension T_1 , the length is L_1 , so the extension is

$$\Delta L_1 = L_1 - L. \text{ Thus, } Y = \frac{T_1 L}{A(L_1 - L)}.$$

For tension T_2 , the length is L_2 , so the extension is

$$\Delta L_2 = L_2 - L. \text{ Thus, } Y = \frac{T_2 L}{A(L_2 - L)}.$$

Equating the two expressions for Y :

$$\frac{T_1 L}{A(L_1 - L)} = \frac{T_2 L}{A(L_2 - L)}$$

$$\frac{T_1}{L_1 - L} = \frac{T_2}{L_2 - L}$$

$$T_1(L_2 - L) = T_2(L_1 - L)$$

$$T_1 L_2 - T_1 L = T_2 L_1 - T_2 L$$

$$T_2 L - T_1 L = T_2 L_1 - T_1 L_2$$

$$L(T_2 - T_1) = T_2 L_1 - T_1 L_2$$

$$L = \frac{T_2 L_1 - T_1 L_2}{T_2 - T_1}$$

- (13) If the linear momentum is increased by 5%, the kinetic energy will increase by.....%

- (A) 5 (B) 10
(C) 10.25 (D) 25

Solution:(Correct Answer:C)

(C) The relationship between kinetic energy E and linear momentum p is given by $E = \frac{p^2}{2m}$.

For small percentage changes, we can use the differential method: $\frac{\Delta E}{E} = 2 \left(\frac{\Delta p}{p} \right)$.

Given $\frac{\Delta p}{p} = 5\% = 0.05$, the change in kinetic energy is

$$\frac{\Delta E}{E} = 2 \times 5\% = 10\%.$$

Alternatively, using the exact formula:

$$E' = \frac{(1.05p)^2}{2m} = 1.1025E.$$

The percentage increase is $(1.1025 - 1) \times 100 = 10.25\%$.

Since 10.25% is the precise value, we select the closest standard approximation or the exact value if provided.

- (14) A monoatomic gas is heated at constant pressure. The percentage of total heat used for doing external work is (in %)

- (A) 30 (B) 40
(C) 50 (D) 60

Solution:(Correct Answer:B)

(B) For a process at constant pressure, the first law of thermodynamics is given by $Q = \Delta U + W$.

The heat supplied is $Q = nC_p \Delta T$.

The change in internal energy is $\Delta U = nC_v \Delta T$.

The external work done is $W = Q - \Delta U =$

$$nC_p \Delta T - nC_v \Delta T = n(C_p - C_v) \Delta T = nR \Delta T.$$

For a monoatomic gas, $C_v = \frac{3}{2}R$ and $C_p = \frac{5}{2}R$.

The fraction of heat used for work is

$$\frac{W}{Q} = \frac{nR \Delta T}{nC_p \Delta T} = \frac{R}{C_p} = \frac{R}{\frac{5}{2}R} = \frac{2}{5}.$$

Converting to percentage: $\frac{2}{5} \times 100\% = 40\%$.

- (15) Identify the pair whose dimensions are equal.

- (A) Torque and work
(B) Stress and energy
(C) Force and stress
(D) Force and work

Solution:(Correct Answer:A)

(A) The dimensional formula for Torque is $[ML^2T^{-2}]$.

The dimensional formula for Work is $[ML^2T^{-2}]$.

Since both have the same dimensional formula, they are equivalent in dimensions.

Therefore, the correct pair is Torque and Work.

- (16) If the degree of freedom of a gas is f , then the ratio of two specific heats C_P/C_V is given by

- (A) $1 + \frac{2}{f}$ (B) $1 - \frac{2}{f}$
(C) $1 + \frac{1}{f}$ (D) $1 - \frac{1}{f}$

Solution:(Correct Answer:A)

(A) The ratio of specific heats is defined as $\gamma = \frac{C_P}{C_V}$.

According to the kinetic theory of gases, the molar specific heat at constant volume is $C_V = \frac{f}{2}R$.

The molar specific heat at constant pressure is

$$C_P = C_V + R = \frac{f}{2}R + R = R(1 + \frac{f}{2}).$$

Therefore, the ratio

$$\gamma = \frac{C_P}{C_V} = \frac{R(1+f/2)}{(f/2)R} = \frac{1+f/2}{f/2} = \frac{2}{f} + 1 = 1 + \frac{2}{f}.$$

- (17) The magnitude of the sum of the two vectors $\vec{A}$ and $\vec{B}$ is equal to the magnitude of the difference of the two vectors $\vec{A}$ and $\vec{B}$. The angle between $\vec{A}$ and $\vec{B}$ is: (in $^\circ$)

- (A) 30 (B) 45
(C) 90 (D) 180

Solution:(Correct Answer:C)

(C) Let the two vectors be $\vec{A}$ and $\vec{B}$ with magnitudes A and B respectively.

The magnitude of their sum is given by:

$$|\vec{A} + \vec{B}| = \sqrt{A^2 + B^2 + 2AB \cos \theta}, \text{ where } \theta \text{ is the angle between the vectors.}$$

The magnitude of their difference is given by:

$$|\vec{A} - \vec{B}| = \sqrt{A^2 + B^2 - 2AB \cos \theta}.$$

Given that $|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}|$, we square both sides:

$$A^2 + B^2 + 2AB \cos \theta = A^2 + B^2 - 2AB \cos \theta.$$

Subtracting $A^2 + B^2$ from both sides, we get:

$$2AB \cos \theta = -2AB \cos \theta.$$

$$4AB \cos \theta = 0.$$

Since A and B are magnitudes of vectors, $A \neq 0$ and $B \neq 0$, therefore $\cos \theta = 0$.

This implies $\theta = 90^\circ$.

- (18) A vehicle travels half the distance with speed v and the remaining distance with speed $2v$. Its average speed is:

- (A) $\frac{3v}{4}$ (B) $\frac{v}{3}$
(C) $\frac{2v}{3}$ (D) $\frac{4v}{3}$

Solution:(Correct Answer:D)

(D) Let the total distance be S .

The vehicle travels the first half distance ($S/2$) with speed $v_1 = v$.

The time taken for the first half is $t_1 = \frac{S/2}{v} = \frac{S}{2v}$.

The vehicle travels the remaining half distance ($S/2$) with speed $v_2 = 2v$.

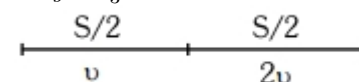
The time taken for the second half is $t_2 = \frac{S/2}{2v} = \frac{S}{4v}$.

The average speed is defined as the total distance divided by the total time:

$$V_{avg} = \frac{\text{Total Distance}}{\text{Total Time}} = \frac{S}{t_1 + t_2}$$

$$V_{avg} = \frac{S}{\frac{S}{2v} + \frac{S}{4v}} = \frac{S}{\frac{2S + S}{4v}} = \frac{S}{\frac{3S}{4v}}$$

$$V_{avg} = \frac{4v}{3}$$



- (19) If the focal length of the objective lens is increased, then the magnifying power of:

- (A) microscope will increase but that of telescope will decrease.
- (B) microscope and telescope both will increase.
- (C) microscope and telescope both will decrease.
- (D) microscope will decrease but that of telescope will increase.

Solution:(Correct Answer:D)

(D) The magnifying power of a compound microscope is given by $m = \left(\frac{L}{f_o}\right) \left(\frac{D}{f_e}\right)$, where f_o is the focal length of the objective lens, f_e is the focal length of the eyepiece, L is the tube length, and D is the least distance of distinct vision. From this formula, it is clear that $m \propto \frac{1}{f_o}$. Therefore, if f_o increases, the magnifying power of the microscope will decrease.

The magnifying power of an astronomical telescope is given by $m = \frac{f_o}{f_e}$, where f_o is the focal length of the objective lens and f_e is the focal length of the eyepiece.

From this formula, it is clear that $m \propto f_o$. Therefore, if f_o increases, the magnifying power of the telescope will increase.

- (20) Silicon and copper are cooled from 300 K to 100 K. What happens to their specific resistance (resistivity)?

- (A) Increase in both copper and silicon
- (B) Decrease in both copper and silicon
- (C) Decrease in copper and increase in silicon
- (D) Increase in copper but decrease in silicon

Solution:(Correct Answer:C)

(C) The resistivity of a conductor (like copper) is directly proportional to temperature. As the temperature decreases, the resistivity of copper decreases.

Conversely, the resistivity of a semiconductor (like silicon) is inversely related to temperature because the number of charge carriers increases with temperature. As the temperature decreases, the number of free charge carriers decreases, causing the resistivity of silicon to increase. Therefore, when cooled from 300 K to 100 K, the resistivity of copper decreases and the resistivity of silicon increases.

- (21) For a purely inductive or a purely capacitive circuit, the power factor is

- (A) zero (B) 0.5
- (C) 1 (D) ∞

Solution:(Correct Answer:A)

(A) The power factor of an AC circuit is defined as $\cos \phi$, where ϕ is the phase difference between the voltage and the current.

In a purely inductive circuit, the current lags behind the voltage by a phase angle of $\phi = 90^\circ$ (or $\pi/2$ radians).

In a purely capacitive circuit, the current leads the voltage by a phase angle of $\phi = 90^\circ$ (or $\pi/2$ radians).

Therefore, the power factor is $\cos(90^\circ) = 0$ for both cases.

- (22) The self-inductance (L) produced by a solenoid of length ' l ' having ' N ' number of turns and cross-sectional area ' A ' is given by the formula (where ϕ = magnetic flux, μ_0 = permeability of vacuum).

- (A) $L = N\phi$ (B) $L = \mu_0 N A l$
- (C) $L = \frac{\mu_0 N^2 A}{l}$ (D) $L = \frac{\mu_0 N A}{l}$

Solution:(Correct Answer:C)

(C) The magnetic field inside the solenoid is given by $B = \frac{\mu_0 N I}{l}$.

The total magnetic flux linkage through the solenoid is

$$\phi_{total} = N(BA).$$

Substituting the value of B , we get

$$\phi_{total} = N \left(\frac{\mu_0 N I}{l} \right) A = \frac{\mu_0 N^2 I A}{l}.$$

By definition, self-inductance L is given by $L = \frac{\phi_{total}}{I}$.

$$\text{Therefore, } L = \frac{\mu_0 N^2 I A}{I} = \frac{\mu_0 N^2 A}{l}.$$

- (23) Choose the correct relation between polarisation P and electric susceptibility χ_e of a dielectric material. (E = electric field)

- (A) $P = \frac{\chi_e}{E^2}$ (B) $P = \frac{\chi_e}{E}$
- (C) $P = \chi_e E$ (D) $P = \chi_e^2 E$

Solution:(Correct Answer:C)

(C) The polarisation P of a dielectric material is defined as the induced dipole moment per unit volume.

For a linear isotropic dielectric, the induced polarisation P is directly proportional to the applied external electric field E . Mathematically, this is expressed as $P = \epsilon_0 \chi_e E$, where ϵ_0 is the permittivity of free space and χ_e is the electric susceptibility.

In many simplified contexts or systems of units where ϵ_0 is absorbed or considered, the relation is given as $P = \chi_e E$. Therefore, the correct relation is $P = \chi_e E$.

- (24) The correct relation between total magnetic field (B), magnetic intensity (H), permeability of free space (μ_0) and susceptibility (χ) is

- (A) $\frac{B}{H} = \mu_0(1 - \chi)$ (B) $\frac{B}{H} = \mu_0(1 + \chi)^2$
- (C) $\frac{B}{H} = \mu_0(1 + \chi)$ (D) $\frac{B}{H} = \mu_0(1 - \chi)^2$

Solution:(Correct Answer:C)

(C) The total magnetic field (B) inside a material is the sum of the magnetic field due to external current ($B_0 = \mu_0 H$) and the magnetic field due to magnetization ($B_m = \mu_0 M$). $B = B_0 + B_m = \mu_0 H + \mu_0 M = \mu_0(H + M)$.

Since the magnetization (M) is related to the magnetic intensity (H) by the susceptibility (χ) as $M = \chi H$, we substitute this into the equation:

$$B = \mu_0(H + \chi H) = \mu_0 H(1 + \chi).$$

Rearranging this gives the ratio:

$$\frac{B}{H} = \mu_0(1 + \chi).$$

- (25) The ratio of contributions made by the electric field and magnetic field components to the intensity of an electromagnetic wave is:

- (A) 1 : c^2 (B) c : 1
- (C) 1 : 1 (D) 1 : c

Solution:(Correct Answer:C)

(C) In an electromagnetic wave (EMW), the energy density associated with the electric field is $u_E = \frac{1}{2} \epsilon_0 E^2$ and the energy density associated with the magnetic field is

$$u_B = \frac{1}{2} \frac{B^2}{\mu_0}.$$

Since $E = cB$ and $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$, we have $u_E = u_B$.

Because the intensity of the wave is proportional to the energy density, the contributions of the electric field and magnetic field components to the total intensity are equal. Therefore, the ratio is 1 : 1.

- (26) Electromagnetic waves are transverse in nature is evident by
- Polarization
 - Interference
 - Reflection
 - Diffraction

Solution:(Correct Answer:A)

(A) The transverse nature of electromagnetic waves is confirmed by the phenomenon of polarization. Polarization is a property that is exclusive to transverse waves, as it involves the restriction of the oscillation of the electric field vector to a specific plane perpendicular to the direction of wave propagation. Longitudinal waves, such as sound waves, cannot be polarized because their oscillations occur along the direction of propagation. Therefore, the correct option is A.

- (27) If ϵ_0 and ϵ represent the permittivity of free space and absolute permittivity of a medium, then the relative permittivity of the medium is:

- $\frac{\epsilon^2}{\epsilon_0}$
- $\frac{\epsilon_0}{\epsilon}$
- $\epsilon\epsilon_0$
- $\frac{\epsilon}{\epsilon_0}$

Solution:(Correct Answer:D)

(D) The absolute permittivity of a medium (ϵ) is related to the permittivity of free space (ϵ_0) and the relative permittivity (ϵ_r) by the formula: $\epsilon = \epsilon_r \epsilon_0$. From this relation, the relative permittivity (ϵ_r) is given by: $\epsilon_r = \frac{\epsilon}{\epsilon_0}$. Therefore, the correct option is D.

- (28) In an interference experiment, the spacing between successive maxima or minima is

- $\lambda d/D$
- $\lambda D/d$
- dD/λ
- $\lambda d/4D$

Solution:(Correct Answer:B)

(B) In an interference experiment, the distance between two consecutive bright fringes (maxima) or two consecutive dark fringes (minima) is known as the fringe width, denoted by β . According to the theory of Young's double-slit experiment, the fringe width is given by the formula: $\beta = \frac{D\lambda}{d}$ where D is the distance between the screen and the slits, λ is the wavelength of the light used, and d is the distance between the two slits.

- (29) A 42 mH inductor is connected to a 200 V, 50 Hz a.c. supply. The r.m.s. value of current in the circuit will be nearly (Take $\pi = \frac{22}{7}$) (in A)

- 15.15
- 9.15
- 8.15
- 6.15

Solution:(Correct Answer:A)

(A) Given: Inductance $L = 42 \text{ mH} = 42 \times 10^{-3} \text{ H}$, Voltage $V_{rms} = 200 \text{ V}$, Frequency $f = 50 \text{ Hz}$. First, calculate the inductive reactance X_L using the formula $X_L = \omega L = 2\pi fL$. Substituting the values: $X_L = 2 \times \frac{22}{7} \times 50 \times 42 \times 10^{-3} \Omega$. $X_L = 2 \times 22 \times 50 \times 6 \times 10^{-3} \Omega = 4400 \times 6 \times 10^{-3} \Omega = 13.2 \Omega$. The r.m.s. current I_{rms} is given by $I_{rms} = \frac{V_{rms}}{X_L}$. $I_{rms} = \frac{200}{13.2} \approx 15.15 \text{ A}$.

- (30) For a thin prism, if the angle of prism is 4° and the refractive index is 1.6, then the angle of minimum deviation will be (in $^\circ$)

- 1.6
- 2.0
- 2.4
- 0.4

Solution:(Correct Answer:C)

(C) For a thin prism, the angle of minimum deviation (D_m) is given by the formula:

$$D_m = A(n - 1)$$

where A is the angle of the prism and n is the refractive index.

Given:

$$A = 4^\circ$$

$$n = 1.6$$

Substituting the values into the formula:

$$D_m = 4^\circ(1.6 - 1)$$

$$D_m = 4^\circ(0.6)$$

$$D_m = 2.4^\circ$$

Therefore, the angle of minimum deviation is 2.4° .

- (31) According to Maxwell's hypothesis, a changing electric field gives rise to

- An e.m.f.
- Electric current
- Magnetic field
- Pressure radiant

Solution:(Correct Answer:C)

(C) According to Maxwell's EM theory, a time-varying electric field produces a displacement current, which in turn acts as a source of a magnetic field.

This is represented by the Maxwell-Ampere law:

$\oint B \cdot dl = \mu_0(I_c + I_d)$, where $I_d = \epsilon_0 \frac{d\Phi_E}{dt}$ is the displacement current.

Therefore, a changing electric field gives rise to a magnetic field.

- (32) Magnetic permeability is maximum for

- Diamagnetic substance
- Paramagnetic substance
- Ferromagnetic substance
- All of these

Solution:(Correct Answer:C)

(C) Magnetic permeability (μ) of a material is defined as its ability to support the formation of a magnetic field within itself when placed in an external magnetic field.

For diamagnetic substances, $\mu < \mu_0$.

For paramagnetic substances, $\mu > \mu_0$.

For ferromagnetic substances, $\mu \gg \mu_0$.

Therefore, magnetic permeability is maximum for ferromagnetic substances.

- (33) An electric kettle has two heating coils. When one of the coils is connected to an a.c. source, the water in the kettle boils in 10 minutes. When the other coil is used, the water boils in 40 minutes. If both the coils are connected in parallel, the time taken by the same quantity of water to boil will be min.

- 15
- 25
- 8
- 4

Solution:(Correct Answer:C)

(C) Let Q be the heat required to boil the water. The heat produced by a coil is given by $Q = \frac{V^2}{R} \times t$, where V is the voltage, R is the resistance, and t is the time.

For the first coil: $Q = \frac{V^2}{R_1} \times t_1$, so $\frac{1}{R_1} = \frac{Q}{V^2 t_1}$.
 For the second coil: $Q = \frac{V^2}{R_2} \times t_2$, so $\frac{1}{R_2} = \frac{Q}{V^2 t_2}$.
 When connected in parallel, the equivalent resistance R_{eq} is given by $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$.
 The heat equation for the parallel combination is $Q = \frac{V^2}{R_{eq}} \times t$, which gives $\frac{1}{R_{eq}} = \frac{Q}{V^2 t}$.
 Substituting these into the parallel resistance formula:
 $\frac{Q}{V^2 t} = \frac{Q}{V^2 t_1} + \frac{Q}{V^2 t_2}$.
 This simplifies to $\frac{1}{t} = \frac{1}{t_1} + \frac{1}{t_2}$.
 Substituting the given values $t_1 = 10$ and $t_2 = 40$:
 $\frac{1}{t} = \frac{1}{10} + \frac{1}{40} = \frac{4+1}{40} = \frac{5}{40} = \frac{1}{8}$.
 Therefore, $t = 8 \text{ min}$.

- (34) Two coherent sources of light interfere and produce a fringe pattern on a screen. For the central maximum, the phase difference between the two waves will be,

(A) $\frac{\pi}{2}$ (B) 0
 (C) π (D) $\frac{3\pi}{2}$

Solution:(Correct Answer:B)

(B) In an interference pattern, the condition for constructive interference is that the path difference Δx must be an integral multiple of the wavelength λ , i.e., $\Delta x = n\lambda$ where $n = 0, 1, 2, \dots$.

The central maximum occurs at the point where the path difference between the two waves is zero ($\Delta x = 0$).

The relationship between phase difference ϕ and path difference Δx is given by $\phi = \frac{2\pi}{\lambda} \Delta x$.

Substituting $\Delta x = 0$ into the equation, we get

$$\phi = \frac{2\pi}{\lambda} \times 0 = 0.$$

Therefore, for the central maximum, the phase difference is 0.

- (35) A charge Q is placed at the centre of a cube of side a . The total flux of electric field through the six surfaces of the cube is

(A) $\frac{6Qa^2}{\epsilon_0}$ (B) $\frac{Qa^2}{6\epsilon_0}$
 (C) Q/ϵ_0 (D) Qa^2/ϵ_0

Solution:(Correct Answer:C)

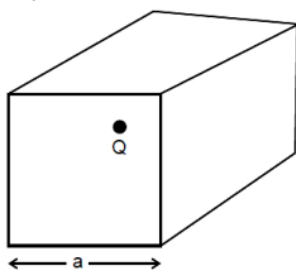
(C) According to Gauss's Law, the total electric flux ϕ through any closed surface is equal to the net charge Q_{enclosed} inside the surface divided by the permittivity of free space ϵ_0 .

Mathematically, $\phi = \frac{Q_{\text{enclosed}}}{\epsilon_0}$.

Since the charge Q is placed at the centre of the cube, the entire charge is enclosed by the cube.

Therefore, the total flux through the six surfaces of the cube is $\phi = \frac{Q}{\epsilon_0}$.

Hint :



- (36) Assertion: The resolving power of a telescope is more if the diameter of the objective lens is more.
 Reason: Objective lens of large diameter collects more light.

- (A) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
 (B) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
 (C) If the Assertion is correct but Reason is incorrect.
 (D) If both the Assertion and Reason are incorrect.

Solution:(Correct Answer:B)

(B) The resolving power of a telescope is defined as the inverse of the smallest angular separation between two distant objects that can be just resolved by the telescope. The formula for the resolving power of a telescope is given by: Resolving Power = $\frac{D}{1.22\lambda}$, where D is the diameter of the objective lens and λ is the wavelength of light used.

From this formula, it is clear that the resolving power is directly proportional to the diameter D of the objective lens. Thus, the Assertion is correct.

The Reason states that an objective lens of large diameter collects more light. This statement is physically true, as a larger aperture allows more light to enter the telescope, which increases its light-gathering power (brightness of the image). However, this is not the reason for the increase in resolving power. The resolving power depends on the diffraction limit, not the light-gathering capacity.

Therefore, both statements are correct, but the Reason is not the correct explanation for the Assertion.

- (37) Radioactive nuclei that are injected into a patient collect at certain sites within their body, undergoing radioactive decay and emitting electromagnetic radiation. These radiations can then be recorded by a detector. This procedure provides an important diagnostic tool called:

(A) Gamma camera
 (B) CAT scan
 (C) Radiotracer technique
 (D) Gamma ray spectroscopy

Solution:(Correct Answer:C)

(C) Radioactive nuclei injected into a patient accumulate at specific sites within the body.

These nuclei undergo radioactive decay and emit electromagnetic radiation (typically gamma rays).

These radiations are detected by external detectors to map the distribution of the tracer.

This diagnostic procedure is known as the Radiotracer technique.

- (38) The refractive index of water is 1.33. What will be the speed of light in water?

(A) $3 \times 10^8 \text{ m/s}$ (B) $2.25 \times 10^8 \text{ m/s}$
 (C) $4 \times 10^8 \text{ m/s}$ (D) $1.33 \times 10^8 \text{ m/s}$

Solution:(Correct Answer:B)

(B) The refractive index μ is defined as the ratio of the speed of light in a vacuum (c) to the speed of light in the medium (v): $\mu = \frac{c}{v}$.

Given: $\mu = 1.33$ and $c = 3 \times 10^8 \text{ m/s}$.

Rearranging the formula to solve for v : $v = \frac{c}{\mu}$.

Substituting the values: $v = \frac{3 \times 10^8}{1.33} \approx 2.25 \times 10^8 \text{ m/s}$.

Therefore, the speed of light in water is $2.25 \times 10^8 \text{ m/s}$.

- (39) A metal surface is illuminated by light of given intensity and frequency to cause photoemission. If the intensity of illumination is reduced to one fourth of its original value, then the maximum KE of the emitted photoelectrons would be

(A) Twice the original value
(B) Four times the original value
(C) One fourth of the original value
(D) Unchanged

Solution:(Correct Answer:D)

(D) The maximum kinetic energy of photoelectrons is given by Einstein's photoelectric equation: $KE_{\max} = h(\nu - \nu_0) \dots$

(i)

Where h is Planck's constant, ν is the frequency of incident radiation, and ν_0 is the threshold frequency.

From equation (i), it is clear that the maximum kinetic energy ($KE_{\max}$) depends only on the frequency of the incident light and the work function of the metal.

It is independent of the intensity of the incident radiation. Therefore, if the intensity is reduced to one-fourth of its original value, the maximum kinetic energy of the emitted photoelectrons remains unchanged.

- (40) Given the value of Rydberg constant is 10^7 m^{-1} , the wave number of the last line of the Balmer series in the hydrogen spectrum will be:

(A) $0.5 \times 10^7 \text{ m}^{-1}$ (B) $0.25 \times 10^7 \text{ m}^{-1}$
(C) $2.5 \times 10^7 \text{ m}^{-1}$ (D) $0.025 \times 10^4 \text{ m}^{-1}$

Solution:(Correct Answer:B)

(B) The wave number $\bar{\nu}$ for the hydrogen spectrum is given by the Rydberg formula: $\bar{\nu} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$.

For the Balmer series, the transition ends at $n_1 = 2$. The last line of the Balmer series corresponds to the transition from $n_2 = \infty$ to $n_1 = 2$.

Given $R = 10^7 \text{ m}^{-1}$, we substitute the values:

$$\bar{\nu} = 10^7 \left(\frac{1}{2^2} - \frac{1}{\infty^2} \right) = 10^7 \left(\frac{1}{4} - 0 \right) = \frac{10^7}{4} = 0.25 \times 10^7 \text{ m}^{-1}.$$

- (41) The current associated with an electron of charge e moving in a circular orbit of radius r with speed v around the positively charged nucleus is given by

(A) $\frac{ev}{2\pi r}$ (B) $\frac{e\pi}{2rv}$
(C) $\frac{ev}{\pi r}$ (D) $\frac{ev}{2\pi r}$

Solution:(Correct Answer:D)

(D) The current i associated with an electron of charge e moving in a circular orbit of radius r with speed v is defined as the rate of flow of charge.

$$i = \frac{q}{T}$$

Here, $q = e$ is the charge of the electron.

The time period T is the time taken to complete one revolution in the circular orbit of circumference $2\pi r$ with speed v .

$$T = \frac{\text{Distance}}{\text{Speed}} = \frac{2\pi r}{v}$$

Substituting the values into the current formula:

$$i = \frac{e}{(2\pi r/v)} = \frac{ev}{2\pi r}$$

- (42) If the initial tension on a stretched string is doubled, then the ratio of the initial and final speeds of a transverse wave along the string is:

(A) $\sqrt{2} : 1$ (B) $1 : \sqrt{2}$
(C) $1 : 2$ (D) $1 : 1$

Solution:(Correct Answer:B)

(B) The speed of a transverse wave on a stretched string is given by the formula $v = \sqrt{\frac{T}{\mu}}$, where T is the tension and μ is the linear mass density.

Since μ remains constant, the speed is directly proportional to the square root of the tension: $v \propto \sqrt{T}$.

Let the initial tension be $T_i = T$ and the final tension be $T_f = 2T$.

The ratio of the initial speed v_i to the final speed v_f is given by:

$$\frac{v_i}{v_f} = \sqrt{\frac{T_i}{T_f}}$$

Substituting the values, we get:

$$\frac{v_i}{v_f} = \sqrt{\frac{T}{2T}} = \sqrt{\frac{1}{2}} = \frac{1}{\sqrt{2}}$$

Thus, the ratio is $1 : \sqrt{2}$.

- (43) If the magnitude of the sum of two vectors is equal to the magnitude of the difference of the two vectors, the angle between these vectors is °.

(A) 90 (B) 120
(C) 45 (D) 60

Solution:(Correct Answer:A)

(A) Let the two vectors be $\vec{A}$ and $\vec{B}$.

The magnitude of the sum of $\vec{A}$ and $\vec{B}$ is given by:

$$|\vec{A} + \vec{B}| = \sqrt{A^2 + B^2 + 2AB \cos \theta}.$$

The magnitude of the difference of $\vec{A}$ and $\vec{B}$ is given by:

$$|\vec{A} - \vec{B}| = \sqrt{A^2 + B^2 - 2AB \cos \theta}.$$

Given that $|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}|$, we square both sides:

$$A^2 + B^2 + 2AB \cos \theta = A^2 + B^2 - 2AB \cos \theta.$$

Subtracting $A^2 + B^2$ from both sides, we get:

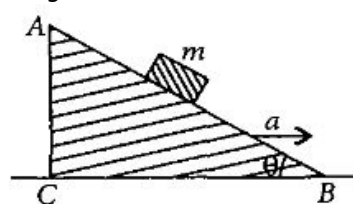
$$2AB \cos \theta = -2AB \cos \theta.$$

$$4AB \cos \theta = 0.$$

Since A and B are non-zero vectors, $4AB \neq 0$, therefore $\cos \theta = 0$.

This implies $\theta = 90^\circ$.

- (44) A block of mass m is placed on a smooth inclined wedge ABC of inclination θ as shown in the figure. The wedge is given an acceleration a towards the right. The relation between a and θ for the block to remain stationary on the wedge is



(A) $a = g \csc \theta$ (B) $a = g \sin \theta$
(C) $a = g \tan \theta$ (D) $a = g \cos \theta$

Solution:(Correct Answer:C)

(C) To keep the block stationary on the wedge, we analyze the forces in the non-inertial frame of the wedge.

1. The forces acting on the block are: gravitational force mg (downwards), normal reaction N (perpendicular to the surface), and pseudo force ma (towards the left).

2. Resolving the forces into components parallel and perpendicular to the inclined surface:

- Perpendicular to the surface: $N = mg \cos \theta + ma \sin \theta$.

However, for the block to be stationary relative to the wedge, the net force along the incline must be zero.

- Parallel to the incline: $ma \cos \theta = mg \sin \theta$.

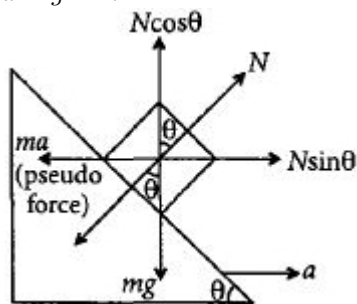
3. From the parallel component equation:

$$ma \cos \theta = mg \sin \theta$$

$$a \cos \theta = g \sin \theta$$

$$a = g \frac{\sin \theta}{\cos \theta}$$

$$a = g \tan \theta$$



- (45) A disc and a sphere of same radius but different masses roll off on two inclined planes of the same altitude and length. Which one of the two objects gets to the bottom of the plane first?

- (A) Sphere
(B) Both reach at the same time
(C) Depends on their masses
(D) Disc

Solution:(Correct Answer:A)

(A) The time t taken by a body to roll down an inclined plane of length l and inclination θ without slipping is given by:

$$t = \sqrt{\frac{2l(1 + \frac{k^2}{R^2})}{g \sin \theta}}$$

where k is the radius of gyration and R is the radius of the object.

Since l, θ , and g are the same for both, the time depends on the factor $(1 + \frac{k^2}{R^2})$.

For a disc, $I = \frac{1}{2}MR^2$, so $k^2 = \frac{1}{2}R^2$, which gives

$$(1 + \frac{k^2}{R^2}) = 1 + 0.5 = 1.5.$$

For a solid sphere, $I = \frac{2}{5}MR^2$, so $k^2 = \frac{2}{5}R^2$, which gives

$$(1 + \frac{k^2}{R^2}) = 1 + 0.4 = 1.4.$$

Since $1.4 < 1.5$, the time taken by the sphere is less than the time taken by the disc ($t_s < t_d$).

Therefore, the sphere reaches the bottom first.

Chemistry - Section A (MCQ)

- (46) What is the vapour density of O_2 gas?

- (A) 8 (B) 16
(C) 32 (D) 22.4

Solution:(Correct Answer:B)

(B) The vapour density of a gas is defined as the ratio of the mass of a certain volume of the gas to the mass of an equal volume of hydrogen gas at the same temperature and pressure.

It is calculated using the formula:

$$\text{Vapour Density} = \frac{\text{Molar Mass of Gas}}{2}$$

The molar mass of O_2 gas is 32 g/mol .

Therefore, Vapour Density = $\frac{32}{2} = 16$.

- (47) Which among the following compounds exhibits +2 oxidation state of oxygen?

- (A) H_2O (B) SO_2
(C) OF_2 (D) H_2O_2

Solution:(Correct Answer:C)

(C) The oxidation number of F is -1 in all of its compounds. Let the oxidation state of O be x .

In OF_2 , the sum of oxidation states is $x + 2(-1) = 0$.

Therefore, $x - 2 = 0$, which gives $x = +2$.

- (48) Which of the following rules or principles states that "No two electrons in an atom can have the same set of four quantum numbers"?

- (A) Pauli's exclusion principle
(B) Hund's rule
(C) Aufbau rule
(D) Heisenberg uncertainty principle

Solution:(Correct Answer:A)

(A) According to Pauli's exclusion principle, no two electrons in an atom can have the same set of four quantum numbers (n, l, m_l, m_s).

This implies that an orbital can hold a maximum of two electrons, and they must have opposite spins.

- (49) Calculate the mass in kg of 4.48 dm^3 of carbon dioxide (CO_2) at STP .

- (A) 2.2×10^{-3} (B) 4.4×10^{-3}
(C) 6.6×10^{-3} (D) 8.8×10^{-3}

Solution:(Correct Answer:D)

(D) At STP , 1 mol of any gas occupies 22.4 dm^3 volume.

The molar mass of CO_2 is $12 + (2 \times 16) = 44 \text{ g/mol}$.

Thus, 22.4 dm^3 of CO_2 at STP has a mass of 44 g .

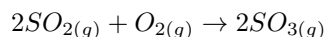
Therefore, the mass of 4.48 dm^3 of CO_2 at STP is calculated as:

$$\text{Mass} = \frac{44 \text{ g}}{22.4 \text{ dm}^3} \times 4.48 \text{ dm}^3 = 8.8 \text{ g}.$$

To convert the mass into kg , we divide by 1000:

$$8.8 \text{ g} = 8.8 \times 10^{-3} \text{ kg}.$$

- (50) For the following reaction, the relation between ΔH and ΔU is:



- (A) $\Delta H = \Delta U - 2RT$ (B) $\Delta H = \Delta U - RT$
(C) $\Delta H = \Delta U + 2RT$ (D) $\Delta H = \Delta U + RT$

Solution:(Correct Answer:B)

(B) The relationship between enthalpy change (ΔH) and internal energy change (ΔU) is given by the equation:

$$\Delta H = \Delta U + \Delta n_g RT.$$

For the reaction: $2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$.

The change in the number of moles of gaseous species is

calculated as: $\Delta n_g = \sum n_{p(g)} - \sum n_{r(g)}$.

$$\Delta n_g = 2 - (2 + 1) = 2 - 3 = -1.$$

Substituting this value into the equation, we get:

$$\Delta H = \Delta U + (-1)RT.$$

Therefore, $\Delta H = \Delta U - RT$.

- (51) What is the number of electrons around sulfur in H_2SO_4 molecule?

- (A) 4 (B) 6
(C) 10 (D) 12

Solution:(Correct Answer:D)

(D) In the H_2SO_4 molecule, the sulfur atom is bonded to four oxygen atoms.

There are two single bonds with two $-OH$ groups and two double bonds with two oxygen atoms.

Each single bond contributes 2 electrons and each double bond contributes 4 electrons to the valence shell of sulfur.

Total electrons around sulfur =

$$(2 \times 2) + (2 \times 4) = 4 + 8 = 12 \text{ electrons}.$$

- (52) The reaction quotient Q for the reaction $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$ is given by $Q = \frac{[NH_3]^2}{[N_2][H_2]^3}$. The reaction will proceed from right to left when:
- (A) $Q = 0$ (B) $Q = K_c$
 (C) $Q < K_c$ (D) $Q > K_c$

Solution:(Correct Answer:D)

(D) The direction of a reaction can be predicted by comparing the reaction quotient Q with the equilibrium constant K_c .
 If $Q > K_c$, the concentration of products is higher than at equilibrium, so the reaction proceeds in the reverse direction (right to left) to reach equilibrium.
 If $Q < K_c$, the reaction proceeds in the forward direction (left to right).
 If $Q = K_c$, the reaction is at equilibrium.
 Therefore, the correct condition for the reaction to proceed from right to left is $Q > K_c$.

- (53) Which of the following atoms should have the highest 1st electron affinity?
- (A) F (B) O
 (C) N (D) C

Solution:(Correct Answer:A)

(A) The electron affinity generally increases across a period from left to right due to an increase in effective nuclear charge and a decrease in atomic size.
 Among the given elements (C, N, O, F), F has the highest effective nuclear charge and the smallest atomic size.
 Although Cl has the highest electron affinity in the entire periodic table, among the provided options, F has the highest 1st electron affinity.

- (54) How many isomers are possible for an alkane having molecular formula C_5H_{12} ?
- (A) 5 (B) 3
 (C) 4 (D) 2

Solution:(Correct Answer:B)

(B) There are 3 possible isomers for an alkane with the molecular formula C_5H_{12} . These are:
 1. n -pentane: $CH_3CH_2CH_2CH_2CH_3$
 2. Isopentane (2-methylbutane): $CH_3 - CH(CH_3) - CH_2 - CH_3$
 3. Neopentane (2,2-dimethylpropane): $C(CH_3)_4$

- (55) Identify the extensive property amongst the following.
- (A) Viscosity
 (B) Heat capacity
 (C) Density
 (D) Surface tension

Solution:(Correct Answer:B)

(B) Extensive properties are those that depend on the amount of matter present in the system.
Heat capacity is an extensive property because it depends on the total amount of substance.
 Viscosity, density, and surface tension are intensive properties as they are independent of the amount of matter present.

- (56) Isomers have essentially identical
- (A) Structural formula
 (B) Chemical properties
 (C) Molecular formula
 (D) Physical properties

Solution:(Correct Answer:C)

(C) Isomers are defined as compounds that possess the same *Molecular formula* but differ in the arrangement of atoms in space or their connectivity.
 Since they share the same *Molecular formula*, they contain the same number and type of atoms.

- (57) Which of the following formulae is used to find the $[OH^-]$ ion concentration of a weak monoacidic base?
- (A) $K_b \cdot c$ (B) $\sqrt{K_b \cdot c}$
 (C) $\sqrt{\frac{K_b}{c}}$ (D) $\sqrt{K_b}$

Solution:(Correct Answer:B)

(B) For a weak monoacidic base BOH , the dissociation equilibrium is represented as: $BOH \rightleftharpoons B^+ + OH^-$.
 Let the initial concentration be c and the degree of dissociation be α .
 The equilibrium concentrations are:
 $[BOH] = c(1 - \alpha)$, $[B^+] = c\alpha$, and $[OH^-] = c\alpha$.
 The dissociation constant K_b is given by:
 $K_b = \frac{[B^+][OH^-]}{[BOH]} = \frac{(c\alpha)(c\alpha)}{c(1-\alpha)} = \frac{c\alpha^2}{1-\alpha}$.
 Since the base is weak, $\alpha \ll 1$, so $1 - \alpha \approx 1$.
 Thus, $K_b \approx c\alpha^2$, which gives $\alpha = \sqrt{\frac{K_b}{c}}$.
 The concentration of $[OH^-]$ is $c\alpha = c \cdot \sqrt{\frac{K_b}{c}} = \sqrt{K_b \cdot c}$.

- (58) Which of the following functional groups has the lowest priority for deciding the principal functional group in a polyfunctional compound?
- (A) $-COOR$ (B) $-COCl$
 (C) $>C=O$ (D) $-CONH_2$

Solution:(Correct Answer:C)

(C) The priority order for functional groups in *IUPAC* nomenclature is as follows:
 $-COOH > -SO_3H > -COOR > -COCl > -CONH_2 > -CN > -CHO > >C=O > -OH > -NH_2 > -C \equiv C- > -C=C-$
 Comparing the given options:
 1. $-COOR$ (Ester)
 2. $-COCl$ (Acyl chloride)
 3. $>C=O$ (Ketone)
 4. $-CONH_2$ (Amide)
 According to the priority order, the ketone group ($>C=O$) has the lowest priority among the given choices.

- (59) How many structural isomers are possible for a compound with molecular formula C_3H_7Cl ?
- (A) 2 (B) 5
 (C) 7 (D) 9

Solution:(Correct Answer:A)

(A) The molecular formula C_3H_7Cl corresponds to a saturated alkyl halide.
 Two structural isomers are possible for this formula:
 1. $CH_3 - CH_2 - CH_2Cl$ (1-chloropropane)
 2. $CH_3 - CHCl - CH_3$ (2-chloropropane)
 Thus, the total number of structural isomers is 2.

(60) During Lassaigne's test, the elements present in an organic compound are converted from :

- (A) ionic form to ionic form
(B) covalent form to ionic form
(C) ionic form to covalent form
(D) covalent form to covalent form

Solution:(Correct Answer:B)

(B) In Lassaigne's test, the organic compound is fused with metallic sodium.

This process converts the covalently bonded elements (like N, S, and Halogens) present in the organic compound into their corresponding water-soluble sodium salts, which are ionic in nature (e.g., $NaCN$, Na_2S , NaX).

(61) In the acetylene molecule, the $C-H$ sigma bond is formed by the overlap of which orbitals?

- (A) $s-p$ overlap (B) $sp-s$ overlap
(C) $p-p$ overlap (D) $sp-sp$ overlap

Solution:(Correct Answer:B)

(B) In acetylene ($HC \equiv CH$), each carbon atom undergoes sp hybridization.

The $C-H$ sigma bond is formed by the head-on overlap of the sp hybrid orbital of the carbon atom and the $1s$ orbital of the hydrogen atom.

Therefore, the bond is formed by $sp-s$ overlap.

(62) What is de Broglie's wavelength for a particle having mass $6.64 \times 10^{-27} \text{ kg}$ moving with velocity of $3 \times 10^3 \text{ ms}^{-1}$ (in nm)? [$h = 6.63 \times 10^{-34} \text{ Js}$]

- (A) 0.111 (B) 0.0333
(C) 1.22 (D) 2.42

Solution:(Correct Answer:B)

(B) The de Broglie wavelength λ is given by the formula $\lambda = \frac{h}{mv}$.

Given: $h = 6.63 \times 10^{-34} \text{ J} \cdot s$, $m = 6.64 \times 10^{-27} \text{ kg}$, and $v = 3 \times 10^3 \text{ ms}^{-1}$.

Substituting the values: $\lambda = \frac{6.63 \times 10^{-34}}{6.64 \times 10^{-27} \times 3 \times 10^3} \text{ m}$.

$\lambda = \frac{6.63}{19.92} \times 10^{-34-(-27)-3} \text{ m}$.

$\lambda \approx 0.333 \times 10^{-10} \text{ m} = 0.333 \times 10^{-9} \text{ m}$.

Since $1 \text{ nm} = 10^{-9} \text{ m}$, we get $\lambda = 0.333 \text{ nm}$.

(63) Which amongst the following is the most stable carbocation?

- (A) $CH_3-CH^+(CH_3)$ (B) $CH_3-C^+(CH_3)_2$
(C) CH_3^+ (D) $CH_3-CH_2^+$

Solution:(Correct Answer:B)

(B) The correct answer is (B) $CH_3-C^+(CH_3)_2$.

Carbocation stability is determined by the inductive effect (+I) and hyperconjugation.

$CH_3-C^+(CH_3)_2$ is a 3° carbocation, which is stabilized by the electron-donating effect of three methyl groups.

$CH_3-CH^+(CH_3)$ is a 2° carbocation, $CH_3-CH_2^+$ is a 1° carbocation, and CH_3^+ is a methyl carbocation.

Stability order: $3^\circ > 2^\circ > 1^\circ > \text{methyl}$.

(64) One atomic mass unit (amu) is equal to:

- (A) $1.66 \times 10^{-27} \text{ g}$ (B) $1.66 \times 10^{-24} \text{ g}$
(C) $1.66 \times 10^{-23} \text{ g}$ (D) $1.66 \times 10^{-25} \text{ kg}$

Solution:(Correct Answer:B)

(B) The atomic mass unit (amu), also known as the unified atomic mass unit (u), is defined as $1/12$ of the mass of a carbon-12 atom.

$1 \text{ amu} = 1.66057 \times 10^{-27} \text{ kg}$.

Since $1 \text{ kg} = 10^3 \text{ g}$, we convert the value to grams:

$1 \text{ amu} = 1.66057 \times 10^{-27} \text{ kg} \times 10^3 \text{ g/kg} = 1.66057 \times 10^{-24} \text{ g}$.
Therefore, the correct value is $1.66 \times 10^{-24} \text{ g}$.

(65) Which of the following is the most stable carbocation?

- (A) CH_3^+ (B) $CH_3CH_2^+$
(C) $(CH_3)_2CH^+$ (D) $(CH_3)_3C^+$

Solution:(Correct Answer:D)

(D) The stability of carbocations is determined by the inductive effect and hyperconjugation.

1. CH_3^+ is a methyl carbocation (zero alkyl groups).

2. $CH_3CH_2^+$ is a primary (1°) carbocation.

3. $(CH_3)_2CH^+$ is a secondary (2°) carbocation.

4. $(CH_3)_3C^+$ is a tertiary (3°) carbocation.

As the number of alkyl groups attached to the positively charged carbon increases, the stability increases due to the electron-donating inductive effect (+I effect) and hyperconjugation.

Therefore, the tertiary carbocation $(CH_3)_3C^+$ is the most stable.

(66) Benzene reacts with CH_3COCl in the presence of $AlCl_3$ to give

- (A) C_6H_5Cl (B) C_6H_5COCl
(C) $C_6H_5CH_3$ (D) $C_6H_5COCH_3$

Solution:(Correct Answer:D)

(D) The reaction of benzene with acetyl chloride (CH_3COCl) in the presence of anhydrous aluminum chloride ($AlCl_3$) is known as the Friedel-Crafts acylation reaction.

In this reaction, the acetyl group ($-COCH_3$) replaces a hydrogen atom on the benzene ring to form acetophenone ($C_6H_5COCH_3$).

The chemical equation is:

$C_6H_6 + CH_3COCl \xrightarrow{AlCl_3} C_6H_5COCH_3 + HCl$.

Therefore, the correct option is (d).

(67) Geometrical isomerism is shown by

- (A) 2-butene (B) 2-butyne
(C) 2-butanol (D) Butanal

Solution:(Correct Answer:A)

(A) Geometrical isomerism is shown by compounds that have restricted rotation around a double bond and where each carbon atom of the double bond is attached to two different groups.

In 2-butene ($CH_3-CH=CH-CH_3$), the carbon atoms of the double bond are each bonded to a hydrogen atom (H) and a methyl group (CH_3).

Thus, it exists as cis-2-butene and trans-2-butene, which are geometrical isomers.

(68) The reagent commonly used to determine the hardness of water titrimetrically is

- (A) Oxalic acid
(B) Disodium salt of EDTA
(C) Sodium citrate
(D) Sodium thiosulphate

Solution:(Correct Answer:B)

(B) The disodium salt of *EDTA* (Ethylenediaminetetraacetic acid) is commonly used in complexometric titrations to determine the hardness of water.

It forms stable, water-soluble complexes with Ca^{2+} and Mg^{2+} ions, which are responsible for water hardness.

- (69) Which of the following enzymes is useful in the conversion of glucose to fructose?

(A) Invertase (B) Glucose isomerase
(C) Amylase (D) Proteases

Solution:(Correct Answer:B)

(B) *Glucose (xylose) isomerase* catalyzes the reversible isomerization of *glucose* to *fructose* and that of *xylose* to *xylulose*.

It is an important enzyme used in the industrial production of high-*fructose* corn syrup.

- (70) Identify the name of the following reaction:

$Toluene + CrO_2Cl_2 \xrightarrow{CS_2} \text{complex} \xrightarrow{H_3O^+} \text{Benzaldehyde}$

(A) Stephen reaction
(B) Rosenmund reaction
(C) Etard reaction
(D) Wolf-Kishner reaction

Solution:(Correct Answer:C)

(C) The reaction of *Toluene* with *Chromylchloride* (CrO_2Cl_2) in the presence of *Carbondisulfide* (CS_2) followed by acid hydrolysis (H_3O^+) to form *Benzaldehyde* is known as the *Etard reaction*.

- (71) Which one of the elements with the following outer orbital configurations may exhibit the largest number of oxidation states?

(A) $3d^5 4s^1$ (B) $3d^5 4s^2$
(C) $3d^2 4s^2$ (D) $3d^3 4s^2$

Solution:(Correct Answer:B)

(B) The number of oxidation states exhibited by a transition element depends on the number of valence electrons available for bonding.

- For $3d^5 4s^1$ (Chromium), the total number of valence electrons is 6, so it can show oxidation states up to +6.
 - For $3d^5 4s^2$ (Manganese), the total number of valence electrons is 7, so it can show oxidation states up to +7.
 - For $3d^2 4s^2$ (Titanium), the total number of valence electrons is 4, so it can show oxidation states up to +4.
 - For $3d^3 4s^2$ (Vanadium), the total number of valence electrons is 5, so it can show oxidation states up to +5.
- Thus, the element with the configuration $3d^5 4s^2$ exhibits the largest number of oxidation states.

- (72) Which one of the following on reduction with lithium aluminium hydride yields a secondary amine?

(A) Methyl isocyanide (B) Acetamide
(C) Methyl cyanide (D) Nitroethane

Solution:(Correct Answer:A)

(A) $CH_3NC + 4[H] \xrightarrow{LiAlH_4} CH_3NHCH_3$
 CH_3NC (Methyl isocyanide) on reduction with lithium aluminium hydride ($LiAlH_4$) yields CH_3NHCH_3 (Dimethylamine), which is a secondary (2°) amine.

In contrast, reduction of alkyl cyanides (RCN) with $LiAlH_4$ yields primary (1°) amines (RCH_2NH_2).

Reduction of acetamide (CH_3CONH_2) yields ethylamine ($CH_3CH_2NH_2$), a primary amine.

Reduction of nitroethane ($CH_3CH_2NO_2$) yields ethylamine ($CH_3CH_2NH_2$), a primary amine.

- (73) The resistance of a conductivity cell filled with 0.1 M KCl solution is $100\ \Omega$ and the conductivity is $1.70 \times 10^{-4}\text{ S cm}^{-1}$. What is the cell constant of the cell (in cm^{-1})?

(A) 1.70 (B) 0.017
(C) 0.058 (D) 0.012

Solution:(Correct Answer:B)

(B) The formula for cell constant (G^*) is given by:

$$G^* = \kappa \times R$$

Where:

- κ is the conductivity (S cm^{-1}),
- R is the resistance (Ω).

Given:

- $\kappa = 1.70 \times 10^{-4}\text{ S cm}^{-1}$,
- $R = 100\ \Omega$.

Calculation:

$$G^* = (1.70 \times 10^{-4}\text{ S cm}^{-1}) \times (100\ \Omega) = 0.017\text{ cm}^{-1}.$$

Therefore, the cell constant is 0.017 cm^{-1} .

Hence, the correct option is B.

- (74) Which among the following colours is obtained in Schiff test of aldehydes?

(A) Blue (B) Green
(C) Magenta (D) Black

Solution:(Correct Answer:C)

(C) When an alcoholic solution of an aldehyde is treated with a few drops of Schiff's reagent, a pink, red, or magenta colour appears.

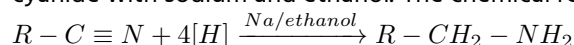
- (75) Alkyl cyanides on reduction by sodium and ethanol give primary amines. This reaction is called as

(A) Wolff-Kishner reduction
(B) Hell-Volhard-Zelinsky reaction
(C) Mendius reduction
(D) Clemmensen reduction

Solution:(Correct Answer:C)

(C)

Primary amines can be obtained by the reduction of alkyl cyanide with sodium and ethanol. The chemical reaction is:



This reaction is known as Mendius reduction.

- (76) A first order reaction takes 40 min for 20% decomposition. Calculate its rate constant.

(A) $5.6 \times 10^{-3}\text{ min}^{-1}$ (B) $4.5 \times 10^{-3}\text{ min}^{-1}$
(C) $6.5 \times 10^{-3}\text{ min}^{-1}$ (D) $7.2 \times 10^{-3}\text{ min}^{-1}$

Solution:(Correct Answer:A)

(A) For a first order reaction, the rate constant k is given by the formula: $k = \frac{2.303}{t} \log_{10} \frac{[A]_0}{[A]_t}$.

Given that the reaction is 20% decomposed, if the initial concentration $[A]_0 = 100$, then the remaining concentration $[A]_t = 100 - 20 = 80$.

Substituting the values into the formula:

$$k = \frac{2.303}{40 \text{ min}} \log_{10} \frac{100}{80}$$

$$k = \frac{2.303}{40} \log_{10}(1.25)$$

Using $\log_{10}(1.25) \approx 0.0969$, we get $k = \frac{2.303 \times 0.0969}{40} \text{ min}^{-1}$.

$$k \approx 0.00557 \text{ min}^{-1} = 5.57 \times 10^{-3} \text{ min}^{-1}$$

Rounding to two significant figures, $k \approx 5.6 \times 10^{-3} \text{ min}^{-1}$.

- (77) The relative lowering of the vapour pressure is equal to the ratio between the number of
- Solute molecules and solvent molecules
 - Solute molecules and the total molecules in the solution
 - Solvent molecules and the total molecules in the solution
 - Solvent molecules and the total number of ions of the solute

Solution:(Correct Answer:B)

(B) According to Raoult's law for a dilute solution containing a non-volatile solute, the relative lowering of vapour pressure is given by the expression:

$$\frac{P^o - P}{P^o} = X_{\text{solute}} = \frac{n}{n+N}$$

Here, n is the number of moles of solute and N is the number of moles of solvent.

Therefore, the relative lowering of vapour pressure is equal to the ratio of the number of solute molecules to the total number of molecules in the solution ($n + N$).

- (78) *EDTA* (Ethylenediaminetetraacetate ion) is a hexadentate ligand. Identify the number of donor atoms in an *EDTA* molecule that form coordinate bonds with a central metal atom or ion in a complex.

- 3
- 1
- 6
- 4

Solution:(Correct Answer:C)

(C) EDTA^{4-} (Ethylenediaminetetraacetate ion) is a hexadentate ligand.

It contains 6 donor atoms: 2 nitrogen atoms (from the amine groups) and 4 oxygen atoms (from the four carboxylate groups).

These 6 donor atoms simultaneously form coordinate bonds with the central metal atom or ion, making it a hexadentate ligand.

Therefore, the number of donor atoms is 6.

- (79) Which catalyst is used in the conversion of chlorobenzene to phenol by the Raschig process?

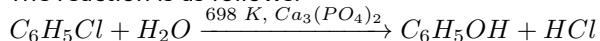
- Calcium phosphate
- Calcium carbonate
- Calcium sulphate
- Calcium chloride

Solution:(Correct Answer:A)

(A) In the Raschig process, chlorobenzene is hydrolyzed to phenol by reacting it with steam at a high temperature of 698 K in the presence of a catalyst.

The catalyst used in this reaction is calcium phosphate, represented by the chemical formula $\text{Ca}_3(\text{PO}_4)_2$.

The reaction is as follows:



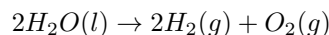
Therefore, the correct option is A.

- (80) During electrolysis of water, the volume of O_2 liberated is 2.24 dm^3 . The volume of hydrogen liberated, under same conditions, will be dm^3 .

- 2.24
- 1.12
- 4.48
- 0.56

Solution:(Correct Answer:C)

(C) The balanced chemical equation for the electrolysis of water is:



According to the stoichiometry of the reaction, 2 moles of H_2 are produced for every 1 mole of O_2 .

Under the same conditions of temperature and pressure, the volume of gas is directly proportional to the number of moles.

Therefore, the volume of H_2 liberated is twice the volume of O_2 liberated.

$$\text{Volume of } \text{H}_2 = 2 \times \text{Volume of } \text{O}_2 = 2 \times 2.24 \text{ dm}^3 = 4.48 \text{ dm}^3$$

- (81) The compound formed as a result of oxidation of ethyl benzene by KMnO_4 is

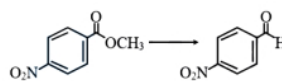
- benzyl alcohol
- benzophenone
- acetophenone
- benzoic acid

Solution:(Correct Answer:D)

(D) When alkyl benzenes (like ethyl benzene) are treated with strong oxidizing agents such as alkaline KMnO_4 , the alkyl side chain is oxidized to a carboxylic acid group ($-\text{COOH}$) attached to the benzene ring, regardless of the length of the alkyl chain, provided that the benzylic carbon has at least one hydrogen atom.

In the case of ethyl benzene ($\text{C}_6\text{H}_5 - \text{CH}_2 - \text{CH}_3$), the benzylic carbon has two hydrogen atoms, so it undergoes oxidation to form benzoic acid ($\text{C}_6\text{H}_5 - \text{COOH}$).

- (82) Identify the reagent used in the following conversion.



- CO , HCl/AlCl_3
- $\text{CrO}_2\text{Cl}_2 \cdot \text{H}_3\text{O}^+$
- $\text{AlH}(\text{i-Bu})_2$, H_3O^+
- CrO_3 , $(\text{CH}_3\text{CO})_2\text{O}$

Solution:(Correct Answer:C)

(C) The given reaction is the reduction of an ester (methyl 4-nitrobenzoate) to an aldehyde (4-nitrobenzaldehyde).

DIBAL-H (Diisobutylaluminium hydride), represented as $\text{AlH}(\text{i-Bu})_2$, is a selective reducing agent that reduces esters to aldehydes at low temperatures, followed by acidic workup (H_3O^+).

Therefore, the correct reagent is $\text{AlH}(\text{i-Bu})_2$, H_3O^+ .

- (83) The instantaneous rate for the reaction $2\text{A} + \text{B} \rightarrow \text{C} + 3\text{D}$ is given by:

- $\frac{1}{3} \frac{d[\text{D}]}{dt}$
- $-\frac{1}{2} \frac{d[\text{A}]}{dt}$
- $-\frac{d[\text{A}]}{dt}$
- $-\frac{d[\text{B}]}{dt}$

Solution:(Correct Answer:A)

(A) For a general reaction $a\text{A} + b\text{B} \rightarrow c\text{C} + d\text{D}$, the rate of reaction is given by $r = -\frac{1}{a} \frac{d[\text{A}]}{dt} = -\frac{1}{b} \frac{d[\text{B}]}{dt} = \frac{1}{c} \frac{d[\text{C}]}{dt} = \frac{1}{d} \frac{d[\text{D}]}{dt}$. Applying this to the reaction $2\text{A} + \text{B} \rightarrow \text{C} + 3\text{D}$, we get:

$$r = -\frac{1}{2} \frac{d[\text{A}]}{dt} = -\frac{d[\text{B}]}{dt} = \frac{d[\text{C}]}{dt} = \frac{1}{3} \frac{d[\text{D}]}{dt}$$

Comparing this with the given options, both A and B are mathematically equivalent expressions for the rate of reaction.

- (84) What is the shape and magnetic nature of the permanganate ion (MnO_4^-)?

- (A) Pyramidal, diamagnetic
 (B) Tetrahedral, diamagnetic
 (C) Tetrahedral, paramagnetic
 (D) Planar, paramagnetic

Solution: (Correct Answer: B)

(B) The permanganate ion (MnO_4^-) has a tetrahedral geometry.
 In this ion, the manganese atom is in the +7 oxidation state, meaning it has a d^0 electronic configuration.
 The π -bonding occurs due to the overlap of p -orbitals of oxygen with d -orbitals of manganese.
 Since there are no unpaired electrons in the d^0 configuration, the ion is diamagnetic.
 The structure is shown below:
 Mn is at the center bonded to four oxygen atoms in a tetrahedral arrangement.

- (85) The conversion of ethyl bromide to ethyl iodide using sodium iodide and dry acetone is known as:

- (A) Swarts reaction
 (B) Finkelstein reaction
 (C) Sandmeyer reaction
 (D) Stephen reaction

Solution: (Correct Answer: B)

(B) The reaction of alkyl halides (like ethyl bromide) with sodium iodide (NaI) in the presence of dry acetone to form alkyl iodides (like ethyl iodide) is a classic example of the Finkelstein reaction.

The reaction is: $C_2H_5Br + NaI \xrightarrow{\text{dry acetone}} C_2H_5I + NaBr$.

- (86) Which of the following electronic configurations represents a transition element?

- (A) $1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^{10}, 4s^2 4p^6$
 (B) $1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^{10}, 4s^2 4p^1$
 (C) $1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^2, 4s^2$
 (D) $1s^2, 2s^2 2p^6, 3s^2 3p^6, 4s^2$

Solution: (Correct Answer: C)

(C) transition element is defined as an element which has an incompletely filled d -orbital in its ground state or in any one of its oxidation states.

Option C represents the configuration

$1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^2, 4s^2$, which corresponds to Titanium ($Ti, Z = 22$).

Since it has a partially filled $3d$ -orbital ($3d^2$), it is a transition element.

- (87) What will be the molality of a solution having 18 g of glucose (mol. wt. = 180) dissolved in 500 g of water?

- (A) 1 (B) 0.5
 (C) 0.2 (D) 2

Solution: (Correct Answer: C)

(C) The formula for molality (m) is given by:

$$m = \frac{\text{mass of solute (g)}}{\text{molar mass of solute (g/mol)} \times \text{mass of solvent (kg)}}$$

Given:

Mass of glucose = 18 g

Molar mass of glucose = 180 g/mol

Mass of water (solvent) = 500 g = 0.5 kg.

Substituting the values:

$$m = \frac{18}{180 \times 0.5} = \frac{0.1}{0.5} = 0.2 m.$$

Therefore, the correct option is (C).

- (88) Identify the product P of the following reaction:

Ethanoyl chloride $\xrightarrow{H_2O}$ P

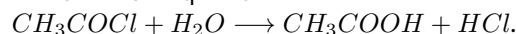
- (A) Ethyl ethanoate (B) Ethanoic acid
 (C) Ethanol (D) Ethanal

Solution: (Correct Answer: B)

(B) The reaction of ethanoyl chloride (CH_3COCl) with water (H_2O) is a hydrolysis reaction.

The chlorine atom is replaced by a hydroxyl group ($-OH$) from water, releasing hydrogen chloride (HCl) as a byproduct.

The chemical equation is:



The product P is ethanoic acid (CH_3COOH).

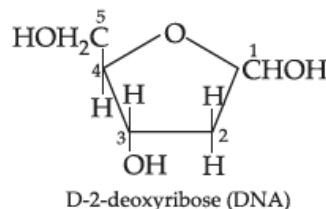
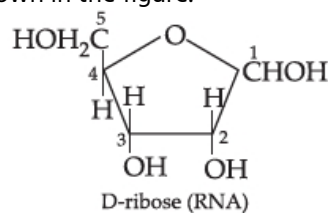
- (89) The presence or absence of a hydroxyl group on which carbon atom of the sugar differentiates RNA and DNA?

- (A) 1st (B) 2nd
 (C) 3rd (D) 4th

Solution: (Correct Answer: B)

(B) The sugar molecule found in RNA is D-ribose, while the sugar in DNA is D-2-deoxyribose.

D-2-deoxyribose differs from ribose only in the substitution of a hydrogen atom for an $-OH$ group at the 2-position, as shown in the figure.



- (90) The magnetic moment (spin only) of $[NiCl_4]^{2-}$ is B.M.

- (A) 1.82 (B) 5.46
 (C) 2.82 (D) 1.41

Solution: (Correct Answer: C)

(C) In $[NiCl_4]^{2-}$, the oxidation state of Ni is +2.

Ni^{2+} has the electronic configuration $[Ar]3d^8$.

Since Cl^- is a weak field ligand, it does not cause pairing of electrons in the $3d$ orbitals.

Thus, there are $n = 2$ unpaired electrons.

The spin-only magnetic moment is calculated as

$$\mu = \sqrt{n(n+2)} \text{ B.M.}$$

$$\mu = \sqrt{2(2+2)} = \sqrt{8} \approx 2.82 \text{ B.M.}$$

Biology - Section A (MCQ)

- (91) How many ATP molecules are formed in one glycolytic pathway?

- (A) 3 (B) 4
 (C) 6 (D) 1

Solution:(Correct Answer:B)

(B) In the glycolytic pathway, 4 molecules of *ATP* are produced through substrate-level phosphorylation. Specifically, 2 molecules of *ATP* are produced during the conversion of 1, 3-bisphosphoglycerate to 3-phosphoglycerate, and 2 molecules of *ATP* are produced during the conversion of phosphoenolpyruvate to pyruvate. Since 2 *ATP* molecules are consumed in the preparatory phase, the net gain is 2 *ATP*, but the total number of *ATP* molecules formed is 4.

- (92) The characteristic feature of brown algae is the presence of

(A) Phycocyanin (B) Phycoerythrin
(C) Fucoxanthin (D) Haematochrome

Solution:(Correct Answer:C)

(C) Brown algae, belonging to the class *Phaeophyceae*, are characterized by the presence of specific pigments that give them their distinct brown color. These algae contain chlorophyll *a, c*, and carotenoids, specifically the xanthophyll pigment known as *Fucoxanthin*. *Fucoxanthin* masks the green color of chlorophyll, resulting in shades ranging from olive green to brown. Phycocyanin and Phycoerythrin are characteristic pigments of red and blue-green algae, respectively.

- (93) During the first meiotic division, the exchange of genetic material between the chromatids of homologous chromosomes occurs. What is this process called?

(A) Transformation
(B) Chiasmata
(C) Crossing over
(D) Synapsis

Solution:(Correct Answer:C)

(C) During the *Pachytene* stage of *Prophase – I* of meiosis, the exchange of genetic material between non-sister chromatids of homologous chromosomes takes place. This process is known as *Crossing over*. *Synapsis* refers to the pairing of homologous chromosomes, while *Chiasmata* are the X-shaped structures formed at the sites where crossing over has occurred.

- (94) Which of the following statements is correct regarding human respiration?

(A) Cigarette smoking causes inflammation of the bronchi.
(B) Neural signals from the pneumotaxic center in the pons region of the brain increase the duration of inspiration.
(C) Workers in grinding and stone-breaking industries suffer from lung fibrosis.
(D) (A) and (C)

Solution:(Correct Answer:D)

(D) 1. Cigarette smoking is a major cause of emphysema and chronic bronchitis, which involves inflammation of the bronchi. Thus, statement (A) is correct.
2. The pneumotaxic center present in the pons region of the brain primarily acts as a 'switch off' point for inspiration. It reduces the duration of inspiration, thereby altering the respiratory rate. Thus, statement (B) is incorrect.

3. In certain industries, especially those involving grinding or stone-breaking, so much dust is produced that the defense mechanism of the body cannot fully cope with the situation. Long-term exposure can give rise to inflammation leading to fibrosis (proliferation of fibrous tissues) and thus cause serious lung damage. Thus, statement (C) is correct.
4. Since both (A) and (C) are correct, the correct option is (D).

- (95) Which one of the following is a matching pair?

(A) Lubb - Sharp closure of *AV* valves at the beginning of ventricular systole.
(B) Dup - Sudden opening of semilunar valves at the beginning of ventricular diastole.
(C) Pulsation of the radial artery valves in the blood vessels.
(D) Purkinje fibres - Initiation of the heart beat.

Solution:(Correct Answer:A)

(A) The first heart sound, *Lubb*, is produced primarily due to the closure of the atrioventricular (*AV*) valves (bicuspid and tricuspid valves) at the beginning of ventricular systole. *Dup* is the second heart sound, caused by the closure of the semilunar valves at the beginning of ventricular diastole. *Purkinje* fibres are responsible for the conduction of the cardiac impulse throughout the ventricles, not for the initiation of the heartbeat (which is the function of the *SA* node).

- (96) Each human kidney has nearly

(A) 10, 000 nephrons (B) 50, 000 nephrons
(C) 1, 00, 000 nephrons (D) 1 million nephrons

Solution:(Correct Answer:D)

(D) The correct answer is *D*. Nephrons are the structural and functional units of the kidney. Each human kidney contains approximately 1 million (10^6) nephrons.

- (97) *ATP* synthesis occurs on the

(A) Outer membrane of mitochondrion
(B) Inner membrane of mitochondrion
(C) Matrix
(D) None of the above

Solution:(Correct Answer:B)

(B) *ATP* synthesis is an endergonic, oxidative process that occurs on the inner mitochondrial membrane, specifically involving the cristae in eukaryotes. In prokaryotes, this process occurs on the inner side of the cell membrane or mesosomes.

- (98) *Caenorhabditiselegans* is a

(A) virus (B) cyanobacterium
(C) fungus (D) nematode

Solution:(Correct Answer:D)

(D) *Caenorhabditiselegans* is a free-living, transparent nematode (roundworm) that is widely used as a model organism in biological research, particularly in developmental biology and genetics. It was the first multicellular organism to have its entire genome sequenced.

(99) Assertion : Hemodialysis can save and prolong the life of uremic patients.
Reason : Waste products like urea can be removed from the blood by the process of hemodialysis.

- (A) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
(B) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
(C) If the Assertion is correct but Reason is incorrect.
(D) If both the Assertion and Reason are incorrect.

Solution:(Correct Answer:A)

(A) In patients suffering from renal failure, the blood urea level rises abnormally, a condition known as uremia. In uremic patients, an artificial kidney (hemodialyzer) is used to remove accumulated nitrogenous waste products like urea from the blood through the process of hemodialysis. Since hemodialysis effectively clears the toxic urea that the damaged kidneys cannot filter, it serves as a life-saving procedure that prolongs the life of these patients. Therefore, the Reason correctly explains why hemodialysis is beneficial for uremic patients.

(100) Vascular cambium and cork cambium are examples of

- (A) Lateral meristem
(B) Apical meristem
(C) Elements of xylem and phloem
(D) Intercalary meristem

Solution:(Correct Answer:A)

(A) Lateral meristems occur laterally in the axis, parallel to the sides of stems and roots. They are responsible for secondary growth, which increases the girth of the plant. The vascular cambium (including fascicular, interfascicular, and extrastelar cambium) and the cork cambium (phellogen) are primary examples of lateral meristems.

(101) Which of the following plants are vascular but seedless?

- (A) Algae (B) Bryophytes
(C) Pteridophytes (D) Gymnosperms

Solution:(Correct Answer:C)

(C) The plant kingdom is classified based on the presence or absence of vascular tissues and seeds.
1. Algae and Bryophytes are non-vascular plants (thallophytes and bryophytes).
2. Pteridophytes are the first terrestrial plants to possess vascular tissues (*xylem* and *phloem*) but they do not produce seeds; they reproduce via spores.
3. Gymnosperms and Angiosperms are vascular plants that produce seeds.
Therefore, Pteridophytes are the correct group of plants that are vascular but seedless.

(102) Synaptonemal complex is formed during

- (A) Meiosis
(B) Amitosis
(C) Mitosis
(D) Cytokinesis

Solution:(Correct Answer:A)

(A) The synaptonemal complex is a protein structure that forms between homologous chromosomes during the *Zygotene* stage of *Prophase – I* of *Meiosis*. It facilitates the process of synapsis, which is the pairing of homologous chromosomes. Therefore, the correct answer is *Meiosis*.

(103) *ECG* helps in the diagnosis of the following abnormalities in conducting pathways *EXCEPT*

- (A) enlargement of heart chambers
(B) damage to cardiac muscles
(C) reduction in blood supply to cardiac muscles
(D) leucopenia

Solution:(Correct Answer:D)

(D) An Electrocardiogram (*ECG*) is a graphical representation of the electrical activity of the heart during a cardiac cycle. It is used to diagnose various heart conditions such as enlargement of heart chambers, damage to cardiac muscles, and reduction in blood supply to cardiac muscles (ischemia). Leucopenia, on the other hand, refers to a decrease in the number of white blood cells (*WBCs*) in the blood, which is a hematological condition and cannot be diagnosed using an *ECG*.

(104) Bulliform cells in grass leaves show

- (A) Growth movements
(B) Tropic movements
(C) Nastic movements
(D) Turgor movements

Solution:(Correct Answer:D)

(D) Bulliform cells are large, bubble-shaped epidermal cells present in the upper epidermis of grass leaves. These cells are thin-walled and contain a large vacuole. When water is abundant, they become turgid, causing the leaf to unfold. When water is scarce, they lose water and become flaccid, causing the leaf to roll inward to reduce the surface area and minimize transpiration. This reversible change in shape due to changes in turgor pressure is known as turgor movement.

(105) All living members of the class Cyclostomata are:

- (A) Free living (B) Endoparasite
(C) Symbiotic (D) Ectoparasite

Solution:(Correct Answer:D)

(D) The class *Cyclostomata* consists of jawless vertebrates. All living members of this class are *Ectoparasites* on some fishes. They possess a circular, sucking mouth without jaws, which they use to attach to the body of host fishes to feed on their blood and flesh. Therefore, the correct option is *D*.

(106) Middle lamella is made up of

- (A) Cellulose
(B) Suberin
(C) Calcium and magnesium pectate
(D) Lignin

Solution:(Correct Answer:C)

(C) The middle lamella is a thin, cementing layer that acts as a glue to hold adjacent plant cells together. It is primarily composed of calcium and magnesium pectates. These substances provide structural stability and adhesion between the cell walls of neighboring cells.

- (107) In which one of the following do male and female gametophytes not have a free-living independent existence?

(A) Polytrichum (B) Cedrus
(C) Pteris (D) Funaria

Solution:(Correct Answer:B)

(B) The correct answer is *B*. In gymnosperms (such as *Cedrus*), the male and female gametophytes do not have an independent, free-living existence. They remain within the sporangia retained on the sporophytes; specifically, the female gametophyte is retained within the megasporangium and the male gametophyte within the microsporangium.

- (108) Match column I with column II for housefly classification and select the correct option using the codes given below.

Column-I	Column-II
(A) Family	(i) Diptera
(B) Order	(ii) Arthropoda
(C) Class	(iii) Muscidae
(D) Phylum	(iv) Insecta

- (A) A – (iii), B – (i), C – (iv), D – (ii)
(B) A – (iii), B – (ii), C – (iv), D – (i)
(C) A – (iv), B – (iii), C – (ii), D – (i)
(D) A – (iv), B – (ii), C – (i), D – (iii)

Solution:(Correct Answer:A)

(A) The taxonomic classification of the housefly (*Musca domestica*) is as follows:

1. Family: Muscidae
2. Order: Diptera
3. Class: Insecta
4. Phylum: Arthropoda

Matching these with the given columns:

- (A) Family corresponds to (iii) Muscidae.
(B) Order corresponds to (i) Diptera.
(C) Class corresponds to (iv) Insecta.
(D) Phylum corresponds to (ii) Arthropoda.

Therefore, the correct sequence is

A – (iii), B – (i), C – (iv), D – (ii).

- (109) Which of the following does *NOT* contribute to the formation of the thoracic cage?

(A) Diaphragm
(B) Sternum
(C) Pleura
(D) Ribs

Solution:(Correct Answer:A)

(A) The thoracic cage is a bony structure that protects the heart and lungs. It is composed of the 12 pairs of ribs, the 12 thoracic vertebrae, and the sternum. The diaphragm is a

muscular partition that separates the thoracic cavity from the abdominal cavity and is not a structural component of the thoracic cage itself. The pleura is a serous membrane that surrounds the lungs, not a part of the skeletal thoracic cage. However, among the given options, the diaphragm is a muscle, while the pleura is a membrane. In the context of skeletal anatomy, the diaphragm is the most distinct non-skeletal structure listed that does not form the cage. Given the standard options, the diaphragm is the primary answer as it forms the floor of the thoracic cavity rather than the cage itself.

- (110) The number of floating ribs in humans is (in pairs)

(A) 2 (B) 3
(C) 5 (D) 6

Solution:(Correct Answer:A)

(A) In the human rib cage, there are 12 pairs of ribs. These are classified into three types based on their attachment to the sternum:

1. True ribs (1st to 7th pairs): These are directly attached to the sternum via hyaline cartilage.
2. False ribs (8th, 9th, and 10th pairs): These do not attach directly to the sternum but join the 7th rib.
3. Floating ribs (11th and 12th pairs): These are not connected to the sternum at all, hence they are called floating ribs.

Therefore, there are 2 pairs of floating ribs.

- (111) What is the response of various organisms to the duration of light and dark periods called?

(A) Phototropism (B) Phototaxis
(C) Vernalization (D) Photoperiodism

Solution:(Correct Answer:D)

(D) The response of plants to periods of day and night (light and dark) is known as *Photoperiodism*.

Phototropism is the growth of an organism in response to a light stimulus.

Phototaxis is the movement of an organism in response to light.

Vernalization is the induction of a plant's flowering process by exposure to the prolonged cold of winter.

- (112) A marine cartilaginous fish that can produce electric current is

(A) Pristis
(B) Torpedo
(C) Trygon
(D) Scoliodon

Solution:(Correct Answer:B)

(B) *Torpedo* is a bottom-living marine cartilaginous fish that can discharge electricity, which is sufficient to stun prey such as small fishes.

A pair of electric organs are situated on the dorsal side of the trunk region.

In fact, these electric organs are modified lateral muscle plates innervated by the cranial nerves.

- (113) Which element is a constituent of all amino acids?

(A) Copper (B) Phosphorus
(C) Nitrogen (D) Magnesium

Solution:(Correct Answer:C)

(C) Amino acids are the building blocks of proteins. Every amino acid has a basic structure consisting of a central carbon atom (alpha-carbon) bonded to an amino group ($-NH_2$), a carboxyl group ($-COOH$), a hydrogen atom, and a variable R -group. Since all amino acids contain an amino group ($-NH_2$), nitrogen is an essential constituent element of all amino acids.

(114) Which of the following statements is incorrect?

- (A) Viroids lack a protein coat.
- (B) Viruses are obligate parasites.
- (C) Infective constituent in viruses is the protein coat.
- (D) Prions consist of abnormally folded proteins.

Solution:(Correct Answer:C)

(C) The correct answer is C .
 1. Viroids consist of free RNA and lack a protein coat, which makes statement A correct.
 2. Viruses are obligate parasites because they require a host cell to replicate, making statement B correct.
 3. In viruses, the genetic material (DNA or RNA) is the infective constituent, not the protein coat (capsid). The protein coat only protects the genetic material. Therefore, statement C is incorrect.
 4. Prions are infectious agents composed entirely of abnormally folded proteins, making statement D correct.

(115) The enzyme required for carboxylation in the Calvin cycle is:

- (A) (1) PEP carboxylase
- (B) (2) RuBP carboxylase-oxygenase
- (C) (3) Carboxypeptidase
- (D) (4) Hexokinase

Solution:(Correct Answer:B)

(B) In the Calvin cycle, the enzyme RuBisCO (RuBP carboxylase-oxygenase) catalyzes the carboxylation step. This enzyme facilitates the fixation of atmospheric CO_2 into a 5-carbon sugar called Ribulose 1, 5-bisphosphate (RuBP).

(116) Given below are two statements:

Statement I : Amino acids have a property of ionizable nature of $-NH_2$ and $-COOH$ groups, hence have different structures at different pH .

Statement II : Amino acids can exist as Zwitterionic form at acidic and basic pH .

In the light of the above statements, choose the most appropriate answer from the options given below:

- (A) Both Statement I and Statement II are correct
- (B) Both Statement I and Statement II are incorrect
- (C) Statement I is correct but Statement II is incorrect
- (D) Statement I is incorrect but Statement II is correct

Solution:(Correct Answer:C)

(C) Statement I is correct. Amino acids are amphoteric because they contain both an acidic carboxyl group ($-COOH$) and a basic amino group ($-NH_2$). These groups can ionize depending on the pH of the solution, leading to different structural forms. Statement II is incorrect. A Zwitterion is a dipolar ion that

has both positive and negative charges, resulting in a net charge of zero. This state occurs at a specific pH known as the isoelectric point (pI). At highly acidic pH , the amino acid exists primarily in a cationic form ($-NH_3^+$), and at highly basic pH , it exists primarily in an anionic form ($-COO^-$). Therefore, it does not exist as a Zwitterion at all acidic or basic pH levels.

(117) A pair of hormones produced by the kidneys is:

- (A) Erythropoietin and Relaxin
- (B) Erythropoietin and Calcitriol
- (C) Calcitonin and Relaxin
- (D) Calcitonin and Calcitriol

Solution:(Correct Answer:B)

(B) The kidneys produce two primary hormones:

1. *Erythropoietin*: This hormone is produced by the interstitial fibroblasts in the kidneys in response to hypoxia (low oxygen levels). It stimulates the production of red blood cells in the bone marrow.
 2. *Calcitriol*: This is the active form of Vitamin D_3 ($1, 25 - dihydroxyvitamin D_3$). It is produced in the kidneys by the enzyme $1\alpha - hydroxylase$ and helps in the absorption of calcium from the gastrointestinal tract. Therefore, the correct pair is *Erythropoietin* and *Calcitriol*.

(118) Singer and Nicholson's model of plasma membrane differs from Robertson's model in the

- (A) Number of lipid layers
- (B) Arrangement of lipid layers
- (C) Arrangement of proteins
- (D) Absence of protein in Singer and Nicholson's model

Solution:(Correct Answer:C)

(C) The fluid mosaic model proposed by Singer and Nicholson (1972) describes the plasma membrane as a mosaic of proteins embedded in a fluid phospholipid bilayer. In contrast, Robertson's unit membrane model proposed that the membrane consists of a phospholipid bilayer sandwiched between two continuous layers of globular proteins.

Therefore, the primary difference between the two models lies in the arrangement of proteins, where Singer and Nicholson suggested that proteins are dispersed (integral and peripheral) within the lipid bilayer rather than forming continuous outer layers.

(119) One very special feature in the earthworm (*Pheretima*) is that

- (A) fertilization of eggs occurs inside the body
- (B) the typhlosole greatly increases the effective absorption area of the digested food in the intestine
- (C) the S -shaped setae embedded in the integument are the defensive weapons used against the enemies
- (D) it has a long dorsal tubular heart.

Solution:(Correct Answer:B)

(B) : In *Pheretima*, the intestine follows the stomach. It is a long, wide, and thin-walled tube extending from the 15th segment to the last. The middle part of the intestine, located between the 27th segment and the 23 – 25 segments in front of the anus, is characterized by the presence of a

highly glandular and vascular longitudinal ridge. This ridge arises as a median ingrowth of the dorsal wall of the intestinal cavity and is called the typhlosole. The typhlosole significantly increases the effective surface area for the absorption of digested food in the intestine.

- (120) The posterior pituitary gland is not a 'true' endocrine gland because

(A) it is provided with a duct
(B) it only stores and releases hormones
(C) it is under the regulation of hypothalamus
(D) it secretes enzymes.

Solution:(Correct Answer:B)

(B) : The posterior lobe of the pituitary gland does not synthesize any hormones. Its hormones, specifically *Oxytocin* and *Vasopressin (ADH)*, are synthesized by the neurosecretory cells of the hypothalamus. The posterior pituitary only stores and releases these hormones into the bloodstream. Therefore, it is not considered a 'true' endocrine gland as it lacks the secretory function characteristic of endocrine tissues.

- (121) The toxic effect of carbon monoxide is due to its greater affinity for haemoglobin as compared to oxygen by (approx.) (in times)

(A) 200 (B) 2
(C) 1000 (D) 20

Solution:(Correct Answer:A)

(A) Carbon monoxide (CO) binds to haemoglobin to form carboxyhaemoglobin ($HbCO$). The affinity of haemoglobin for CO is approximately 200 to 250 times greater than its affinity for oxygen (O_2). Due to this high affinity, CO effectively displaces oxygen from haemoglobin, leading to hypoxia and tissue damage.

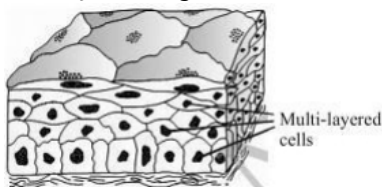
- (122) $K_{2+2}C_4A_{2+4}G_{(2)}$ is the floral formula of:

(A) *Allium cepa* (B) *Solanum nigrum*
(C) *Helianthus annuus* (D) *Brassica nigra*

Solution:(Correct Answer:D)

(D) The floral formula $K_{2+2}C_4A_{2+4}G_{(2)}$ represents the family *Brassicaceae* (formerly *Cruciferae*). K_{2+2} indicates four sepals arranged in two whorls of two each. C_4 indicates four petals arranged in a cruciform manner. A_{2+4} indicates six stamens arranged in two whorls (tetradynamous condition). $G_{(2)}$ indicates a bicarpellary, syncarpous, superior ovary. Among the given options, *Brassica nigra* (Black mustard) belongs to the family *Brassicaceae*.

- (123) Identify the diagram and its related function.



- (A) Simple cuboidal epithelium - Diffusion
(B) Simple squamous epithelium - Secretion and absorption
(C) Compound epithelium - Protection
(D) Compound epithelium - Diffusion

Solution:(Correct Answer:C)

(C) The provided diagram shows a tissue composed of multiple layers of cells, which is characteristic of compound epithelium. Compound epithelium is made of more than one layer (multi-layered) of cells and thus has a limited role in secretion and absorption. Its main function is to provide protection against chemical and mechanical stresses.

- (124) Viruses have

(A) *DNA* enclosed in a protein coat
(B) prokaryotic nucleus
(C) single chromosome
(D) both *DNA* and *RNA*

Solution:(Correct Answer:A)

(A) The correct answer is *A*. Viruses are non-cellular, nucleoprotein entities that utilize the synthetic machinery of a living host cell for their replication. They do not possess a cellular structure, such as a nucleus or organelles. Their genetic material consists of either *DNA* or *RNA*, but never both. This genetic material is enclosed within a protective protein coat called a capsid.

- (125) *ICBN* stands for

(A) International Code of Botanical Nomenclature
(B) International Congress of Biological Names
(C) Indian Code of Botanical Nomenclature
(D) Indian Congress of Biological Names

Solution:(Correct Answer:A)

(A) *ICBN* stands for International Code of Botanical Nomenclature. It is a set of rules and recommendations dealing with the formal botanical names given to plants. The foundations of *ICBN* were established in the book written by *C. Linnaeus* named *Philosophia Botanica*. It is independent of zoological nomenclature. The rank of species is basic, and the relative order of the ranks of taxa is: species, genus, tribe, family, order, series, class, division, and kingdom. The different ranks or categories have specific endings for their names, such as division (-phyta), class (-ae), and family (-aceae).

- (126) Which one of the following types of reaction occurs during the conversion of isocitrate $\rightarrow$ oxalosuccinate?

(A) Oxidation (B) Decarboxylation
(C) Hydration (D) Dephosphorylation

Solution:(Correct Answer:A)

(A) In the citric acid cycle (Krebs cycle), the conversion of isocitrate to oxalosuccinate is catalyzed by the enzyme isocitrate dehydrogenase.

This reaction involves the removal of hydrogen atoms from the substrate, which are transferred to NAD^+ to form $NADH + H^+$.

Since the process involves the removal of hydrogen (a form of oxidation), this reaction is classified as an oxidation reaction.

(127) In *Lathyrus aphaca*, the leaves are modified into

- (A) Spine
- (B) Tendril
- (C) Scale
- (D) Stem-like structure

Solution:(Correct Answer:B)

(B) The correct answer is (b).

In *Lathyrus aphaca* (Wild pea), the entire leaf is modified into slender, wiry, and coiled structures known as leaf tendrils.

These tendrils help the plant in climbing by providing support to the weak stem.

(128) The movement of auxin is largely

- (A) Basipetal
- (B) Acropetal
- (C) Centripetal
- (D) Centrifugal

Solution:(Correct Answer:A)

(A) Auxin exhibits polar transport. In the stem, the movement is primarily basipetal (from the apex towards the base), whereas in the root, it is acropetal (from the root tip towards the shoot). Therefore, the most characteristic and widely recognized movement of auxin in the plant body is basipetal.

(129) Multipolar nerve cells are present in

- (A) Cochlea
- (B) Dorsal root ganglia of spinal cord
- (C) Retina of eye
- (D) Brain

Solution:(Correct Answer:D)

(D) Multipolar neurons are characterized by having one axon and two or more dendrites. They are the most common type of neurons in the vertebrate nervous system and are found in the cerebral cortex of the brain and the spinal cord. In contrast, bipolar neurons are found in the retina of the eye, and pseudounipolar neurons are found in the dorsal root ganglia of the spinal cord.

(130) Which of the following is considered the structural and functional unit of neural tissue?

- (A) Myelin sheath
- (B) Axon
- (C) Dendrite
- (D) Neuron

Solution:(Correct Answer:D)

(D) The neural tissue is composed of neurons and neuroglial cells.

Neurons are the structural and functional units of the nervous system.

They are specialized cells that receive, process, and transmit information through electrical and chemical signals.

While axons, dendrites, and myelin sheaths are parts of a neuron, the entire cell itself is the fundamental unit.

(131) Which of the following statements is wrong for viroids?

- (A) They cause infections.
- (B) Their *RNA* is of high molecular weight.
- (C) They lack a protein coat.
- (D) They are smaller than viruses.

Solution:(Correct Answer:B)

(B) The correct answer is B. Viroids are infectious agents that consist only of a short strand of circular single-stranded *RNA* without a protein coat. A key characteristic of viroid *RNA* is that it has a low molecular weight compared to the *RNA* found in viruses. Therefore, the statement that their *RNA* is of high molecular weight is incorrect.

(132) Who proposed the Law of Limiting Factors?

- (A) Blackman
- (B) Hill
- (C) Arnon
- (D) Liebig

Solution:(Correct Answer:A)

(A) The Law of Limiting Factors was proposed by *F.F. Blackman* in 1905.

According to this law, if a chemical process is affected by more than one factor, then its rate will be determined by the factor which is nearest to its minimal value, and this factor is known as the limiting factor.

(133) Mast cells of connective tissue contain

- (A) Vasopressin and relaxin
- (B) Heparin and histamine
- (C) Heparin and calcitonin
- (D) Serotonin and melanin

Solution:(Correct Answer:B)

(B) Mast cells are specialized cells found in the connective tissue that play a crucial role in the immune response. They contain secretory granules filled with chemical mediators.

The primary substances released by mast cells include heparin (an anticoagulant), histamine (a vasodilator involved in inflammatory responses), and serotonin.

Therefore, the correct combination is heparin and histamine.

(134) The innermost layer of the cortex which shows Casparian thickenings in its cells is called as

- (A) Epidermis
- (B) Endodermis
- (C) Pericycle
- (D) Exodermis

Solution:(Correct Answer:B)

(B) The correct answer is B.

The *Endodermis* is the innermost layer of the cortex in roots.

It consists of a single layer of barrel-shaped, compactly

arranged parenchymatous cells.

The radial and tangential walls of these cells contain water-impermeable, waxy material called *Suberin*, which forms the *Casparian* strips.

(135) In the gel electrophoresis method

- (A) Fragments of *DNA* can be seen without being stained.
- (B) A natural polymer extracted from seaweed is used as the medium (gel).
- (C) Glowing orange bands of *DNA* can be seen upon exposure to visible light on an ethidium bromide-stained gel.
- (D) The larger the size of the *DNA* fragment in the agarose medium, the farther it will move.

Solution:(Correct Answer:B)

(B) In gel electrophoresis, agarose, which is a natural polymer extracted from seaweeds (like *Gelidium* and *Gracilaria*), is used as the matrix or gel medium.

Option A is incorrect because *DNA* fragments are invisible without staining.

Option C is incorrect because *DNA* bands stained with ethidium bromide only glow under *UV* light, not visible light.

Option D is incorrect because, in gel electrophoresis, smaller *DNA* fragments move faster and farther through the agarose matrix than larger fragments due to the sieving effect.

(136) Natural selection where more individuals acquire specific character value other than the mean character value, leads to:

- (A) Directional change (B) Disruptive change
- (C) Random change (D) Stabilising change

Solution:(Correct Answer:A)

(A) In natural selection, when more individuals acquire a value other than the mean character value, the population shifts towards one extreme. This is known as Directional selection or Directional change.

1. Stabilising selection: Individuals with mean character values are favoured.

2. Directional selection: Individuals at one extreme of the character range are favoured, causing the mean to shift.

3. Disruptive selection: Individuals at both extremes of the character range are favoured, leading to two peaks in the distribution.

(137) Which of the following sexually transmitted diseases do not specifically affect reproductive organs?

- (A) Genital warts and Hepatitis-B
- (B) Syphilis and Genital herpes
- (C) *AIDS* and Hepatitis-B
- (D) Chlamydia and *AIDS*

Solution:(Correct Answer:C)

(C) Sexually Transmitted Diseases (*STDs*) are infections transmitted through sexual contact.

While many *STDs* primarily affect the reproductive tract (e.g., Syphilis, Genital herpes, Chlamydia, and Genital warts), some infections transmitted sexually have systemic effects or target other organs.

AIDS (Acquired Immuno Deficiency Syndrome) is caused by the *HIV* virus, which attacks the immune

system, specifically *CD4+ T*-cells, rather than the reproductive organs directly.

Hepatitis-B is a viral infection that primarily affects the liver. Therefore, both *AIDS* and Hepatitis-B are sexually transmitted but do not specifically target the reproductive organs.

(138) What is antisense technology?

- (A) When a piece of *RNA* that is complementary in sequence is used to stop expression of a specific gene
- (B) *RNA* polymerase producing *DNA*
- (C) A cell displaying a foreign antigen used for synthesis of antigens
- (D) Production of somaclonal variants in tissue cultures

Solution:(Correct Answer:A)

(A) *RNA* interference (*RNAi*) is a method of cellular defense in all eukaryotic organisms. This method involves silencing of a specific *mRNA* due to a complementary *dsRNA* molecule that binds to and prevents translation of the *mRNA* (silencing).

1. The *mRNA* produced by a gene is called the sense strand because it contains the codons for translation.

2. A complementary *RNA* strand, which cannot be translated into a functional protein, is called antisense *RNA*.

3. When antisense *RNA* is introduced, it base-pairs with the target *mRNA* to form a double-stranded *RNA* (*dsRNA*) molecule.

4. This *dsRNA* cannot be translated by the ribosome, effectively blocking the expression of the specific gene. This is the basis of antisense technology.

(139) Select the hormone-releasing Intra-Uterine Devices (IUDs).

- (A) Vaults, *LNG – 20*
- (B) Multiload 375, Progestasert
- (C) Progestasert, *LNG – 20*
- (D) Lippes Loop, Multiload 375

Solution:(Correct Answer:C)

(C) Intra-Uterine Devices (IUDs) are effective contraceptive methods.

They are categorized into three types:

1. Non-medicated IUDs: e.g., Lippes Loop.

2. Copper-releasing IUDs: e.g., Copper *T* (*CuT*), Copper 7 (*Cu7*), Multiload 375.

3. Hormone-releasing IUDs: These release hormones that make the uterus unsuitable for implantation and the cervix hostile to the sperms. Examples include Progestasert and *LNG – 20*.

Therefore, the correct option is C.

(140) A population in a habitat with limited resources shows four phases of growth. Identify the correct sequence of growth.

- (A) Acceleration → deceleration → Lag phase → asymptote
- (B) Lag phase → acceleration → deceleration → asymptote
- (C) Lag phase → asymptote → deceleration → Acceleration → lag phase
- (D) Acceleration → lag phase → deceleration → asymptote

Solution:(Correct Answer:B)

(B) In a habitat with limited resources, population growth follows a logistic growth model (Sigmoid curve).

1. Lag phase: The initial phase where the population adapts to the environment and growth is slow.
2. Acceleration phase: The population size increases rapidly as resources are abundant.
3. Deceleration phase: As resources become limited, the growth rate begins to slow down.
4. Asymptote: The population density reaches the carrying capacity (K) of the environment, and growth stabilizes.

- (141) Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R .
 Assertion A : Endometrium is necessary for implantation of blastocyst.
 Reason R : In the absence of fertilization, the corpus luteum degenerates that causes disintegration of endometrium.
 In the light of the above statements, choose the correct answer from the options given below:

- (A) A is false but R is true.
 (B) Both A and R are true and R is the correct explanation of A .
 (C) Both A and R are true but R is NOT the correct explanation of A .
 (D) A is true but R is false.

Solution:(Correct Answer:C)

(C) Both Assertion A and Reason R are true, but R is not the correct explanation of A .

Assertion A is true because the endometrium is the inner lining of the uterus where the blastocyst implants to establish pregnancy.

Reason R is also true because the corpus luteum secretes progesterone, which maintains the endometrium. If fertilization does not occur, the corpus luteum degenerates, leading to a drop in progesterone levels, which causes the endometrium to disintegrate (menstruation). However, the reason R explains why the endometrium is shed during the menstrual cycle, not why the endometrium is necessary for implantation. Therefore, R is not the correct explanation for A .

- (142) Mother and father both have blood group 'A'. They have two children, one with blood group 'O' and the second one with blood group 'A'. They have:

- (A) mother has homozygous gene, father has heterozygote $I^A I^A$.
 (B) both are homozygous ($I^A I^A$).
 (C) mother is heterozygous ($I^A i$) and father is homozygous ($I^A I^A$).
 (D) both are heterozygous ($I^A i$).

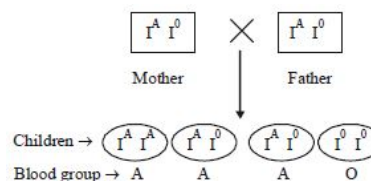
Solution:(Correct Answer:D)

(D) The inheritance of blood groups is determined by multiple alleles. For a child to have blood group 'O' (genotype ii), both parents must contribute an 'i' allele. Since both parents have blood group 'A', their genotypes must be $I^A I^A$ or $I^A i$.

If a child has blood group 'O' (ii), it implies that both parents must carry the recessive 'i' allele.

Therefore, both parents must be heterozygous for blood group 'A', with the genotype $I^A i$.

When both parents are $I^A i$, the possible genotypes for the offspring are $I^A I^A$ (blood group A), $I^A i$ (blood group A), and ii (blood group O).



- (143) Haemophilia is caused due to the lack of:

- (A) ADH (B) AHF
 (C) STH (D) $ACTH$

Solution:(Correct Answer:B)

(B) Haemophilia is a sex-linked recessive genetic disorder in which the blood fails to clot properly.

It is caused by a deficiency of the Anti-Haemophilic Factor (AHF), also known as Factor $VIII$, which is essential for the blood coagulation cascade.

Therefore, the correct option is B .

- (144) Which of the following is not a desirable feature of a cloning vector?

- (A) Presence of a marker gene
 (B) Presence of single restriction enzyme site
 (C) Presence of two or more recognition sites
 (D) Presence of origin of replication

Solution:(Correct Answer:C)

(C) cloning vector must have an origin of replication (ori) to allow autonomous replication.

It should contain a selectable marker gene to identify transformants.

It must have a single recognition site for a specific restriction enzyme to allow the insertion of the foreign DNA .

Having two or more recognition sites for the same restriction enzyme would result in the fragmentation of the vector, which is undesirable as it complicates the cloning process.

- (145) When two species with different genetic backgrounds show similarities due to adaptation, this phenomenon is called:

- (A) Divergent evolution
 (B) Microevolution
 (C) Co-evolution
 (D) Convergent evolution

Solution:(Correct Answer:D)

(D) When two or more unrelated or distantly related species evolve similar traits as a result of having to adapt to similar environments or ecological niches, this process is known as *Convergent evolution*.

This leads to the development of analogous organs, which perform similar functions but have different evolutionary origins.

For example, the wings of butterflies and birds, or the flippers of dolphins and penguins, show convergent evolution.

- (146) When glucose is the normal energy source in a bacterial cell, which gene transcribes a repressor mRNA?

- (A) z (B) y
 (C) a (D) i

Solution:(Correct Answer:D)

(D) In the lac operon model, the i gene is the regulatory gene that codes for the repressor protein.

This repressor protein binds to the operator region to

prevent transcription of the structural genes (z, y, a) when lactose is absent.

Even when glucose is the primary energy source, the i gene is constitutively expressed to produce the repressor mRNA, which is then translated into the repressor protein. Therefore, the i gene is responsible for transcribing the repressor mRNA.

(147) The alleles are

- (A) A pair of genes governing a specific character such as tallness or dwarfness
- (B) Multiple forms of genes
- (C) Genes governing eye characters
- (D) Genes present in allosomes

Solution:(Correct Answer:A)

(A) Alleles are slightly different versions of the same gene that occupy the same locus on homologous chromosomes. They govern the same trait (e.g., height) but may result in different expressions (e.g., tallness or dwarfness). Therefore, they are defined as a pair of genes governing a specific character.

(148) Which was the first successful cloned animal that gained worldwide fame?

- (A) Molly (Sheep) (B) Polly (Sheep)
- (C) Chance (Bull) (D) Dolly (Sheep)

Solution:(Correct Answer:D)

(D) Dolly the sheep was the first mammal to be cloned from an adult somatic cell. It was created by researchers at the Roslin Institute in Scotland and was born on 5 July 1996. This achievement gained worldwide fame as it demonstrated that a specialized cell from an adult animal could be used to create an exact genetic copy of the entire organism.

(149) The function of copper ions in copper-releasing IUDs is:

- (A) they inhibit gametogenesis
- (B) they make the uterus unsuitable for implantation
- (C) they inhibit ovulation
- (D) they suppress sperm motility and the fertilising capacity of sperms

Solution:(Correct Answer:D)

(D) : Copper-releasing IUDs (e.g., CuT , $Cu7$, *Multiload 375*) are placed in the uterus of females. These devices are highly efficient methods of birth control. The copper ions (Cu^{2+}) released by these devices increase phagocytosis of sperms within the uterus. Furthermore, these ions suppress the motility of sperms and reduce their fertilising capacity, thereby preventing fertilization.

(150) What is the approximate concentration of CO_2 in the atmosphere (in %)?

- (A) 0.0314 (B) 0.34
- (C) 3.34 (D) 4.0

Solution:(Correct Answer:A)

(A) The concentration of carbon dioxide (CO_2) in the Earth's atmosphere is approximately 0.03% to 0.04%. Among the given options, 0.0314% is the most accurate

representation of the standard atmospheric concentration of CO_2 (often cited as approximately 314 ppm in historical contexts or standard textbook references). Therefore, option A is the correct answer.

(151) Which animal has become extinct in India?

- (A) Snow leopard (B) Hippopotamus
- (C) Wolf (D) Cheetah

Solution:(Correct Answer:D)

(D) The *Cheetah* (*Acinonyx jubatus*) was declared extinct in India in 1952 due to habitat loss and over-hunting. Although recent efforts have been made to reintroduce them, historically, they are classified as extinct in the wild within the Indian subcontinent.

(152) Gel electrophoresis is used for:

- (A) Construction of recombinant DNA by joining with cloning vectors
- (B) Isolation of DNA molecules
- (C) Cutting of DNA into smaller fragments
- (D) Separation of DNA fragments according to their size

Solution:(Correct Answer:D)

(D) Gel electrophoresis is a technique used to separate charged molecules like DNA, RNA, or proteins based on their size and charge. In the context of DNA technology, it is specifically used to separate DNA fragments based on their size. Under an electric field, DNA fragments (which are negatively charged) move towards the anode through a matrix (usually agarose gel). Smaller fragments move faster and travel further through the gel pores compared to larger fragments, allowing for effective separation.

(153) In *lac* operon, z gene codes for:

- (A) β -galactosidase (B) Permease
- (C) Repressor (D) Transacetylase

Solution:(Correct Answer:A)

(A) In the *lac* operon, the structural genes are z, y , and a .
1. The z gene codes for β -galactosidase, which is responsible for the hydrolysis of lactose into galactose and glucose.
2. The y gene codes for permease, which increases the permeability of the cell to β -galactosides.
3. The a gene codes for transacetylase.
Therefore, the z gene codes for β -galactosidase.

(154) In the life cycle of *Plasmodium vivax*, if a (Trophozoite), b (gametocyte), c (oocyst), d (zygote), e (sporozoite) are different stages. Select the correct sequence starting from the stage it enters the human body.

- (A) $e \rightarrow a \rightarrow b \rightarrow c \rightarrow d$ (B) $e \rightarrow a \rightarrow b \rightarrow d \rightarrow c$
- (C) $c \rightarrow a \rightarrow d \rightarrow b \rightarrow e$ (D) $c \rightarrow b \rightarrow e \rightarrow a \rightarrow d$

Solution:(Correct Answer:B)

(B) The life cycle of *Plasmodium vivax* begins in humans when the infectious stage, the e (sporozoite), enters the human body through the bite of an infected female *Anopheles* mosquito. Inside the human body, the sporozoites infect liver cells and then red blood cells, where they transform into the a (Trophozoite) stage. After asexual reproduction, some parasites differentiate into sexual stages known as b (gametocytes).

When a mosquito bites an infected person, these gametocytes are ingested. In the mosquito's gut, they fuse to form a *d* (zygote).

The zygote then develops into an *c* (oocyst) within the mosquito's gut wall, which eventually releases sporozoites. Therefore, the sequence starting from the stage entering the human body is $e \rightarrow a \rightarrow b \rightarrow d \rightarrow c$.

(155) Match the following.

<i>I</i>	<i>II</i>	<i>III</i>
(<i>P</i>) <i>Trichoderma polysporum</i>	(<i>a</i>) statins	(<i>i</i>) clot buster
(<i>Q</i>) <i>Monascus purpureus</i>	(<i>b</i>) streptokinase	(<i>ii</i>) organ transplant
(<i>R</i>) <i>Streptococcus</i>	(<i>c</i>) cyclosporin-A	(<i>iii</i>) cholesterol lowering agent

(A) $P - (c - iii), Q - (b - i), R - (a - ii)$

(B) $P - (a - iii), Q - (c - i), R - (b - ii)$

(C) $P - (b - ii), Q - (a - i), R - (c - iii)$

(D) $P - (c - ii), Q - (a - iii), R - (b - i)$

Solution:(Correct Answer:D)

(D) The correct matches are as follows:

1. *Trichoderma polysporum* produces cyclosporin-A, which is used as an immunosuppressive agent in organ transplant patients. Thus, $P - (c - ii)$.

2. *Monascus purpureus* produces statins, which act as blood cholesterol-lowering agents. Thus, $Q - (a - iii)$.

3. *Streptococcus* produces streptokinase, which is used as a 'clot buster' for removing clots from the blood vessels of patients who have undergone myocardial infarction. Thus, $R - (b - i)$.

Therefore, the correct sequence is

$P - (c - ii), Q - (a - iii), R - (b - i)$.

(156) The process of removal of anther from the flower bud before it dehisces is called as

(A) Emasculation

(B) Bagging

(C) Embryo rescue

(D) Budding

Solution:(Correct Answer:A)

(A) Emasculation is a technique used in plant breeding to prevent self-pollination.

In this process, the anthers are removed from the bisexual flower bud using a pair of forceps before they dehisce (release pollen).

This ensures that the stigma of the flower is only pollinated by the desired pollen grains from a selected male parent.

(157) Select the *CORRECT* statement.

(A) Darwin's finches living on the Galapagos islands are different in only color from those found on main land.

(B) Darwin's finches living on Galapagos island are different in only vegetarian food from those found on main land.

(C) Darwin's finches living on Galapagos island are different in only shapes of beak.

(D) Darwin's finches living on Galapagos island are different in size, colour, beak and food habits from those on main land.

Solution:(Correct Answer:D)

(D) Darwin's finches are a classic example of adaptive radiation. When Charles Darwin visited the Galapagos Islands, he observed a variety of small black birds, now known as Darwin's finches. He noticed that these birds exhibited significant variations in their size, plumage (colour), beak structure, and feeding habits compared to their ancestral species on the mainland. These variations allowed them to exploit different ecological niches on the islands, such as eating seeds, insects, or cactus fruits. Therefore, option D is the correct statement as it encompasses the diverse phenotypic differences observed.

(158) In human beings, multiple genes are involved in the inheritance of

(A) Colour blindness

(B) Phenylketonuria

(C) Sickle-cell anaemia

(D) Skin colour

Solution:(Correct Answer:D)

(D) Polygenic inheritance is a type of inheritance in which a trait is controlled by two or more genes.

In humans, skin colour is a classic example of polygenic inheritance.

It is influenced by the cumulative effect of multiple genes, resulting in a wide range of phenotypes rather than distinct categories.

In contrast, colour blindness is an X-linked recessive trait, while phenylketonuria and sickle-cell anaemia are examples of monogenic (single-gene) disorders.

(159) Among the following sets of examples for divergent evolution, select the incorrect option:

(A) Eye of octopus, bat and man

(B) Forelimbs of man, bat and cheetah

(C) Brain of bat, man and cheetah

(D) Heart of bat, man and cheetah

Solution:(Correct Answer:A)

(A) Divergent evolution occurs when species with a common ancestor evolve different traits to adapt to different environments, resulting in homologous organs. Homologous organs are those that have the same structural origin and developmental pattern but may perform different functions.

A. The eye of an octopus (a mollusc) and the eye of a mammal (bat or man) are examples of convergent evolution, not divergent evolution. They perform similar functions but have different evolutionary origins.

B. The forelimbs of man, bat, and cheetah are classic examples of homologous organs, showing divergent evolution.

C. The brains of vertebrates like bats, humans, and cheetahs share a common structural plan, indicating divergent

evolution.

D. The hearts of vertebrates like bats, humans, and cheetahs share a common structural plan, indicating divergent evolution.

Therefore, the incorrect option regarding divergent evolution is the eye of an octopus, bat, and man.

- (160) Human population in India reached the 1 billion mark by the year

(A) 1995 (B) 1999
(C) 2000 (D) 2011

Solution:(Correct Answer:C)

(C) The population in India, which was approximately 350 million at the time of independence, reached close to the 1 billion mark by the year 2000 and crossed 1.2 billion in May 2011.

- (161) What is the study of the interrelationships between living organisms and their environment called?

(A) Ecology
(B) Ecosystem
(C) Phytogeography
(D) Bioecology

Solution:(Correct Answer:A)

(A) The study of the interrelationships between living organisms and their environment is known as *Ecology*. *Ecology* is derived from the Greek words *Oikos* (meaning house or dwelling) and *Logos* (meaning study). An *Ecosystem* refers to the functional unit of nature where living organisms interact among themselves and with the surrounding physical environment. *Phytogeography* is the study of the geographic distribution of plants. Therefore, the correct term for the study of these interrelationships is *Ecology*.

- (162) Select the incorrect statement with respect to acquired immunity.

(A) Anamnestic response is elicited on subsequent encounters with the same pathogen.
(B) Anamnestic response is due to memory of first encounter.
(C) Acquired immunity is non-specific type of defense present at the time of birth.
(D) Primary response is produced when our body encounters a pathogen for the first time.

Solution:(Correct Answer:C)

(C) Acquired immunity is pathogen-specific and is characterized by memory. It is not present at birth but develops during an individual's lifetime after exposure to pathogens or vaccines. In contrast, innate immunity is non-specific and is present at the time of birth. Therefore, the statement 'Acquired immunity is non-specific type of defense present at the time of birth' is incorrect.

- (163) Which of the following prevents self-pollination?

(A) Self-sterility
(B) Herkogamy
(C) Dichogamy
(D) All of the above

Solution:(Correct Answer:D)

(D) Self-pollination is prevented by several mechanisms in plants to promote cross-pollination:

1. *Self – sterility* (or self-incompatibility): A genetic mechanism where pollen grains fail to germinate on the stigma of the same flower.
 2. *Herkogamy*: A physical or mechanical barrier between the anther and the stigma of the same flower, preventing self-pollination.
 3. *Dichogamy*: A condition where the anthers and stigma of the same flower mature at different times (protandry or protogyny).
- Therefore, all these mechanisms prevent self-pollination.

- (164) The theory of natural selection is based on which of the following?

(A) Contribution of the environment to evolution
(B) Natural selection acts on favorable variations
(C) Genetic variations result from changes in the gene pool
(D) None of the above

Solution:(Correct Answer:B)

(B) The theory of natural selection, proposed by *Charles Darwin*, is primarily based on the observation that individuals within a population exhibit variations. Some of these variations are heritable and provide a survival or reproductive advantage in a given environment. Natural selection acts on these favorable variations, allowing individuals possessing them to survive and reproduce more successfully than others. Over generations, this leads to the accumulation of these advantageous traits in the population, which is the core mechanism of evolution.

- (165) Hybridomas are employed for

(A) Synthesis of antibiotics
(B) Killing cancer cells
(C) Synthesis of monoclonal antibodies
(D) Production of somatic hybrids

Solution:(Correct Answer:C)

(C) Hybridoma technology is a method used for the production of monoclonal antibodies. In this process, a specific antibody-producing *B*-lymphocyte is fused with a myeloma (cancer) cell to form a hybridoma cell. These hybridoma cells are immortal and can be cultured to produce large quantities of specific, uniform antibodies known as monoclonal antibodies, which are highly specific to a single antigen.

- (166) Which one of the following cells secretes a hormone?

(A) Cells of Leydig
(B) Cells of Sertoli
(C) Primary spermatocyte
(D) Secondary spermatocyte

Solution:(Correct Answer:A)

(A) Leydig cells, also known as interstitial cells, are located in the interstitial spaces between the seminiferous tubules. These cells synthesize and secrete testicular hormones,

primarily *Testosterone*, under the influence of *Luteinizing Hormone (LH)* from the anterior pituitary gland.

- (167) Match the ecological terms given in Column I with its correct explanation in Column II.

Column I (Terms)	Column II (Explanation)
1. Stratification	a. Horizontal distribution of different species
2. Zonation	b. Rate of generation of Biomass in an ecosystem
3. Productivity	c. Breakdown of detritus
4. Decomposition	d. Vertical distribution of different species

Choose the

correct option.

- (A) 1 – d, 2 – c, 3 – a, 4 – b (B) 1 – d, 2 – a, 3 – b, 4 – c
(C) 1 – c, 2 – d, 3 – a, 4 – b (D) 1 – c, 2 – a, 3 – d, 4 – b

Solution:(Correct Answer:B)

(B) 1. Stratification refers to the vertical distribution of different species occupying different levels in an ecosystem.
2. Zonation refers to the horizontal distribution of different species in an ecosystem.
3. Productivity is defined as the rate of generation of biomass in an ecosystem.
4. Decomposition is the process of breakdown of complex organic matter (detritus) into simpler inorganic substances. Therefore, the correct matching is 1 – d, 2 – a, 3 – b, 4 – c.

- (168) In an ecosystem, 1000 J of energy is stored at the producer level. How much energy will be available at the secondary consumer level (in J)?

- (A) 10 (B) 0.1
(C) 100 (D) 1

Solution:(Correct Answer:A)

(A) According to the 10% law of energy transfer in ecosystems, only about 10% of the energy is transferred from one trophic level to the next.
Step 1: Energy at the producer level = 1000 J.
Step 2: Energy at the primary consumer level = 10% of 1000 J = 100 J.
Step 3: Energy at the secondary consumer level = 10% of 100 J = 10 J.
Therefore, the energy available at the secondary consumer level is 10 J.

- (169) One of the following is not a method of contraception. Which one?

- (A) Condoms
(B) Pills of a combination of oxytocin and vasopressin
(C) Lippes loop
(D) Tubectomy

Solution:(Correct Answer:B)

(B) : Oxytocin is a birth hormone, and vasopressin (anti-diuretic hormone) is responsible for reabsorbing water from the renal tubules to conserve water in the body. Neither of these hormones has any role in contraception.

- (170) The yield of a paddy field can be increased by the application of:
(A) Iron bacteria (B) *Nostoc/Anabaena*
(C) Archaeobacteria (D) Symbiotic bacteria

Solution:(Correct Answer:B)

(B) *Nostoc* and *Anabaena* are free-living nitrogen-fixing cyanobacteria. They increase the nitrogen content of moist soil and water bodies. This enrichment of nitrogen acts as a natural fertilizer, which significantly increases the yield of paddy crops.

- (171) A complex of ribosomes attached to a single strand of RNA is known as
(A) polypeptide (B) Okazaki fragment
(C) polysome (D) polymer

Solution:(Correct Answer:C)

(C) During the process of translation, multiple ribosomes can attach to a single mRNA molecule to synthesize multiple copies of the same polypeptide chain simultaneously. This structure, consisting of a single mRNA strand with several ribosomes attached to it, is called a polyribosome or polysome.

- (172) In case of snakebites, the injection given to patients contains preformed antibodies against the snake venom. This type of immunisation is called
(A) Active immunity
(B) Both kinds of immunity
(C) Passive immunity
(D) Partial passive immunity

Solution:(Correct Answer:C)

(C) When ready-made or preformed antibodies are directly injected into the body to provide a quick immune response, it is known as passive immunity. In the case of snakebites, the patient needs immediate protection because the venom acts rapidly. Since the body does not have enough time to produce its own antibodies (which would be active immunity), preformed antibodies are administered to neutralize the toxin immediately. Therefore, the correct answer is C (Passive immunity).

- (173) Which of the following layers in an antral follicle is acellular?
(A) Stroma (B) Zona pellucida
(C) Granulosa (D) Theca interna

Solution:(Correct Answer:B)

(B) The *Zona pellucida* is a thick, transparent, and acellular glycoprotein layer that surrounds the plasma membrane of an oocyte. It is secreted by both the oocyte and the surrounding follicular cells. Other layers like the *Granulosa* cells and *Theca interna* are composed of living cells, while the *Stroma* refers to the connective tissue framework of the ovary.

- (174) In large cities, domestic sewage

- (A) has high *BOD* as it contains both aerobic and anaerobic bacteria.
- (B) is processed in sewage treatment plants through primary aerobic and then secondary anaerobic processes.
- (C) does not require aeration when processed in *STP* because it contains sufficient oxygen.
- (D) contains large amounts of suspended solids and dissolved salts.

Solution:(Correct Answer:A)

(A) Domestic sewage in large cities is characterized by high organic content.

1. *BOD* (Biochemical Oxygen Demand) is a measure of the organic matter present in water.
2. Domestic sewage contains a large amount of biodegradable organic matter, which serves as food for microorganisms like aerobic and anaerobic bacteria.
3. As these microorganisms decompose the organic matter, they consume dissolved oxygen, leading to a high *BOD* value.
4. Therefore, domestic sewage is characterized by high *BOD* due to the presence of high organic load and associated microbial activity.

- (175) In relation to Gross primary productivity and Net primary productivity of an ecosystem, which one of the following statements is correct?
- (A) There is no relationship between Gross primary productivity and Net primary productivity.
 - (B) Gross primary productivity is always less than net primary productivity.
 - (C) Gross primary productivity is always more than net primary productivity.
 - (D) Gross primary productivity and Net primary productivity are one and the same.

Solution:(Correct Answer:C)

(C) Gross primary productivity (*GPP*) is the total amount of organic matter produced by producers through photosynthesis in an ecosystem per unit area over a time period.

Net primary productivity (*NPP*) is the amount of biomass available for consumption by heterotrophs after accounting for the energy lost by producers during respiration (*R*).

The relationship is defined by the equation:

$$NPP = GPP - R.$$

Since respiration (*R*) is always a positive value, *GPP* must always be greater than *NPP*.

- (176) Unisexuality of flowers prevents
- (A) geitonogamy, but not xenogamy
 - (B) autogamy and geitonogamy
 - (C) autogamy, but not geitonogamy
 - (D) both geitonogamy and xenogamy.

Solution:(Correct Answer:C)

(C) : Unisexuality or dicliny is a condition in which two types of unisexual flowers are present, *i.e.*, staminate (male flower) and pistillate (female flower).

In a monoecious plant, unisexual flowers prevent autogamy (self-pollination within the same flower) because the male

and female reproductive parts are in different flowers. However, geitonogamy (transfer of pollen from the anther of one flower to the stigma of another flower on the same plant) can still occur in monoecious plants. Therefore, unisexuality prevents autogamy but does not prevent geitonogamy.

- (177) Which of the following is incorrect regarding vasectomy?

- (A) No sperm occurs in seminal fluid
- (B) No sperm occurs in epididymis
- (C) Vasa deferentia is cut and tied
- (D) Irreversible sterility

Solution:(Correct Answer:B)

(B) : Vasectomy is a surgical contraception method performed in males. In vasectomy, a small part of the vas deferens is removed or tied up through a small cut on the scrotum. This prevents sperm transport. Vasectomy has a poor reversibility. There is no effect on libido and erectile functioning. Seminal vesicles are one pair of sac-like structures which join vasa deferentia to form the ejaculatory duct. They secrete seminal fluid which contains fructose, prostaglandins, and clotting proteins, but no sperms. In a male who has undergone vasectomy, the ejaculatory duct will receive seminal fluid, but due to the cut in vasa deferentia, sperms will not be transported from the epididymis; hence, the semen will lack sperms.

- (178) What is true about cleavage in the fertilised egg in humans?

- (A) It is meroblastic.
- (B) It starts while the egg is in the fallopian tube.
- (C) It is identical to normal mitosis.
- (D) It starts when the egg reaches the uterus.

Solution:(Correct Answer:B)

(B) . Cleavage in the fertilized egg (zygote) in humans is holoblastic and occurs in the fallopian tube (oviduct) as the zygote moves towards the uterus. Unlike normal mitosis, cleavage involves rapid cell divisions without significant growth between divisions, resulting in smaller cells called blastomeres.

- (179) Triploid plants can be obtained from the culture of:

- (A) Pollen
- (B) Endosperm
- (C) Ovule
- (D) Megaspore

Solution:(Correct Answer:B)

(B) In angiosperms, the endosperm is formed as a result of double fertilization, specifically through the fusion of one male gamete with two polar nuclei (triple fusion). This process results in a triploid ($3n$) tissue. Since the endosperm cells are naturally triploid, culturing these cells through tissue culture techniques can lead to the development of triploid plants.

- (180) Which of the following is incorrectly matched regarding the microbe, its derivative, and its application?

- (A) *Monascus purpureus* - Statins - Lowers blood cholesterol levels.
- (B) *Streptococcus* - Streptokinase - Removes clots from blood vessels.
- (C) *Clostridium butylicum* - Lipase - Used to remove oily stains in laundry.
- (D) *Trichoderma polysporum* - Cyclosporin-A - Immunosuppressive agent in organ transplant patients.

Solution:(Correct Answer:C)

(C) The correct answer is C.

1. *Monascus purpureus* produces statins, which act as blood-cholesterol-lowering agents.
2. *Streptococcus* produces streptokinase, which is used as a 'clot buster' for removing clots from the blood vessels of patients who have undergone myocardial infarction.
3. *Clostridium butylicum* is a bacterium that produces butyric acid. Lipase is produced by the fungus *Candida lipolytica* or *Rhizopus arrhizus*, not *Clostridium butylicum*. Lipases are used in detergent formulations and are helpful in removing oily stains from the laundry.
4. *Trichoderma polysporum* is a fungus that produces cyclosporin-A, which is used as an immunosuppressive agent in organ transplant patients.

Vedclass